Low-Level Design – TFA Assistant

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Low-Level Design (sub-module 08.1 Assistant, code `TFA`) · **Version:** 1.0 · **Status:** Draft
> **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026
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> **Based on:** HLD-Assistant v1.0, FRD-Assistant v1.0, TDD v2.0 (§16)
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Document Control

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version: `AssistantController` (7 actions), chat engine, model gateway and providers, 9 tools, pending actions, help knowledge, audit and rate limit, voice input, 4 tables, tests and FR-TFA traceability

Related documents: BRD (`../../../BRD/08-TFA-Assistant/08.1-Assistant/BRD-Assistant.md`) · FRD (`../../../FRD/08-TFA-Assistant/08.1-Assistant/FRD-Assistant.md`) · HLD (`../../../HLD/08-TFA-Assistant/08.1-Assistant/HLD-Assistant.md`) · Overall LLD (`../../00-Overall/LLD-TaskFlow-Overall.md`) · TFA Settings LLD (`../../06-Settings/06.3-TFA-Settings/LLD-TFA-Settings.md`) · TDD (`TF - Documents/TF - TDD/TDD-TaskFlow.md` §16)

1. Introduction

1.1 Purpose

Detailed design of TFA (TaskFlow Assistant) as built: the classes and files of the chat API (`/api/v1/assistant`), the conversation engine with its bounded tool loop, the provider abstraction for Ollama, Gemini and OpenAI, the 9 model-callable tools, the pending-action lifecycle (prepare → confirm once), the embedded help knowledge, the audit log and per-user rate limit, the Angular TFA Panel and its voice input, the four TFA tables, configuration and tests. Every `FR-TFA-nn` of the FRD is traced to an endpoint or class and a test in §11.

1.2 Scope & Conventions

- Child modules: **08.1.1 Chat** (4 actions), **08.1.2 Actions and Tools** (3 actions + 9 tools), **08.1.3 Help Knowledge** (0 actions), **08.1.4 Audit Log and Rate Limits** (0 actions), **08.1.5 Voice Input** (frontend only). Total: **7 controller actions**, all on `AssistantController` (facts.md §3, rows `AssistantController`).

- Out of scope: provider, model and API key configuration – `AiProvidersController` (`/api/v1/assistant/providers/config`, `/api/v1/assistant/providers/{kind}/models`, `/api/v1/assistant/providers/{kind}/test`), `IAiProviderConfigService`, tables `AiProviderConfigs`, `AiModels` (TFA Settings LLD). TFA only reads them.
- Naming: `*Controller`, `I*Service` / `*Service`, `I*Repository` / `*Repository`, tools `*Tool` implementing `IAssistantTool`, DTOs `*Request` / `*Dto` / `*Payload`.
- API base path `/api/v1`; JSON camelCase, enums as strings; errors as `application/problem+json`.
- Paths are relative to `backend/src/TaskFlow.Api/` and `frontend/src/app/`.

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2. Code Structure

Shared layout and layer rules: overall LLD (`../../00-Overall/LLD-TaskFlow-Overall.md`). TFA part only:

2.1 Backend Folder Layout

```
TaskFlow.Api/
  Controllers/Assistant/AssistantController.cs
  Services/Assistant/
    Access/AssistantCaller.cs              (AssistantCaller, AssistantPolicies)
    Chat/IAssistantService.cs, AssistantService.cs, AssistantPrompt.cs, AssistantTexts.cs,
ProviderFailureMessages.cs
    ModelGateway/IChatModelGateway.cs, ChatModelGateway.cs
    Toolbox/IAssistantToolbox.cs, AssistantToolbox.cs (also AssistantToolGate), ToolOutcome.cs
    Tools/IAssistantTool.cs, AssistantReadTools.cs, AssistantActionTools.cs
    Actions/IAssistantActionService.cs, IAssistantActionCards.cs, AssistantActionService.cs
    HelpKnowledge/IHelpKnowledgeBase.cs, HelpKnowledgeBase.cs, HelpTopicDto.cs
    Knowledge/*.md                         (19 embedded topics)
    AuditLog/IAssistantAuditLog.cs, AssistantAuditLog.cs
    RateLimits/IAssistantRateLimiter.cs, AssistantRateLimiter.cs
    Rules/AssistantRules.cs, AssistantActionRules.cs, SecretScreen.cs, TaskPrioritiser.cs
  Repositories/Assistant/Conversations/, Messages/, PendingActions/, People/, Audits/
  Integrations/Ai/Providers/ (IChatModelProvider, ChatModelTypes, AiProviderRuntime,
ProviderFailure, ProviderHttp,
                             OllamaProvider, GeminiProvider, OpenAiProvider)
  Integrations/Ai/Registry/ (IChatModelProviderRegistry, ChatModelProviderRegistry)
  Integrations/Ai/KeyProtection/ (IAiKeyProtector, AiKeyProtector, KeyRingProtectionCheck)
  Authorization/Assistant/IAssistantAuthorizer.cs, AssistantAuthorizer.cs
  Dtos/Assistant/AssistantDtos.cs, AssistantActionDtos.cs, AssistantFailureDto.cs
  Models/Entities/AssistantConversation.cs, AssistantMessage.cs, AssistantPendingAction.cs,
AssistantAudit.cs
  Models/Enums/AssistantActionKind.cs, AssistantActionStatus.cs, AssistantMessageRole.cs,
AiProviderKind.cs, ProviderFailureKind.cs
  Common/Exceptions/AssistantExceptions.cs, Common/Constants/AiProviderCatalog.cs
  Config/ServiceRegistration.cs, RepositoryRegistration.cs, IntegrationRegistration.cs,
AiProviderOptions.cs
```

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2.2 Frontend Folder Layout

```
src/app/modules/assistant/
  components/tfa-panel, tfa-header, tfa-composer, tfa-message, tfa-action-card
(.component.ts|html)
  services/assistant.service.ts, assistant.model.ts, wake-phrase.service.ts, wake-
phrase.model.ts,
           speak.service.ts, speak.model.ts
  models/assistant.model.ts
src/app/layouts/shell.component.ts|html     (@if (canUseAssistant()) renders app-tfa-panel)
src/app/shared/auth/permissions.ts          (AssistantChat, AssistantActions)
```

No route and no menu entry exist for TFA; the panel is part of the shell.

2.3 Layer Rules

Rule	Enforced by
Controller → Service → Repository only; `AssistantController` calls `IAssistantService`, `IAssistantActionService`, `IAiProviderConfigService`	`ArchitectureTests`
Tools reach other modules only through their services (`ITaskService`, `IMailService`, `IProfileService`, `IEmployeeService`)	`ArchitectureTests` (layer rule); code review
HTTP to AI providers only from `Integrations/Ai/`	Folder rule (overall LLD)
Frontend feature folders	`npm run check:structure`

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3. Cross-Cutting Design

Concern	Design
Validation	No FluentValidation class for chat; length and secret checks in `AssistantService`; action edits reuse `SendMailRequestValidator`; profile actions reuse `UpdateOwnProfileRequestValidator`
Error handling	`GlobalExceptionHandler` maps `AppException` to status, title and `detail`; `RetryAfter` sets `Retry-After`. Provider failures are not exceptions to the client: 200 with a `failure` object
Transactions	`IUnitOfWork.SaveChangesAsync` after a successful turn; pending-action claim and status changes use single `ExecuteUpdateAsync` statements
Authorization	Class `[Authorize]` + `perm:{key}` policies per action; tools and confirms re-check the same policies through `IAssistantAuthorizer`
Logging	Provider failures logged with provider, kind and HTTP status only; provider `HttpClient`s have logging handlers removed
Time & time zone	UTC storage; "today" and overdue from `IAppClock` (application time zone, fallback `Application:Timezone`)

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3.1 Exception → HTTP Status

Exception	Status	When (TFA)
`AssistantRequestException` (: `AppException`, title "TFA could not handle that")	400	Message length, no or unknown provider, edit of a non-mail action, invalid mail edit
`AssistantRateLimitedException` (: `AssistantRequestException`, title "Too many requests")	429 + `Retry-After`	More than 20 messages per minute
`AssistantActionConflictException` (: `ConflictException`, title "This action can no longer be done")	409	Expired, already done, no longer pending
– (`NotFound()` from `RunActionAsync`)	404	Action unknown or not the caller's
– (policy)	403	Missing `assistant.chat` / `assistant.actions.confirm`; recorded by `DeniedActionAuditHandler`
`ProviderNotConfiguredException` (not an `AppException`)	500	Provider disabled between availability check and call (see §10)

4. Module Designs

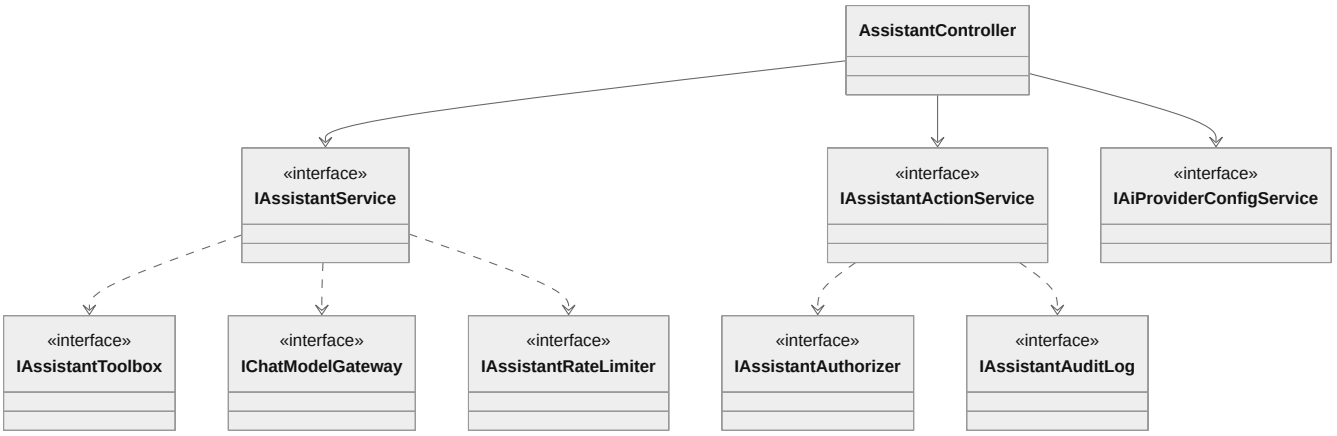
4.1 Module: TFA Assistant (`TFA`)

`AssistantController` (`[Route("api/v1/assistant")]`, `[Authorize]`) has **7 actions**: `GetConversation`, `SendMessage`, `NewConversation`, `GetProviders` (Chat) and `EditAction`, `ConfirmAction`, `CancelAction` (Actions). Every action builds `new AssistantCaller(User)` – user id from `ClaimTypes.NameIdentifier`, role from `ClaimTypes.Role`.

DI (`Config/ServiceRegistration.cs`): singletons `IAssistantRateLimiter` → `AssistantRateLimiter`, `IHelpKnowledgeBase` → `HelpKnowledgeBase`; scoped `IAssistantAuthorizer` → `AssistantAuthorizer`, `AssistantTaskReader`, `AssistantUserResolver`, `AssistantManagerResolver`, `AssistantToolGate`, nine `IAssistantTool` registrations, `IAssistantToolbox` → `AssistantToolbox`, `IAssistantAuditLog` → `AssistantAuditLog`, `IAssistantActionService` → `AssistantActionService` (also resolved as `IAssistantActionCards`), `IAssistantService` → `AssistantService`.

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Module class diagram



4.1.1 Child module: Chat

Class / file inventory

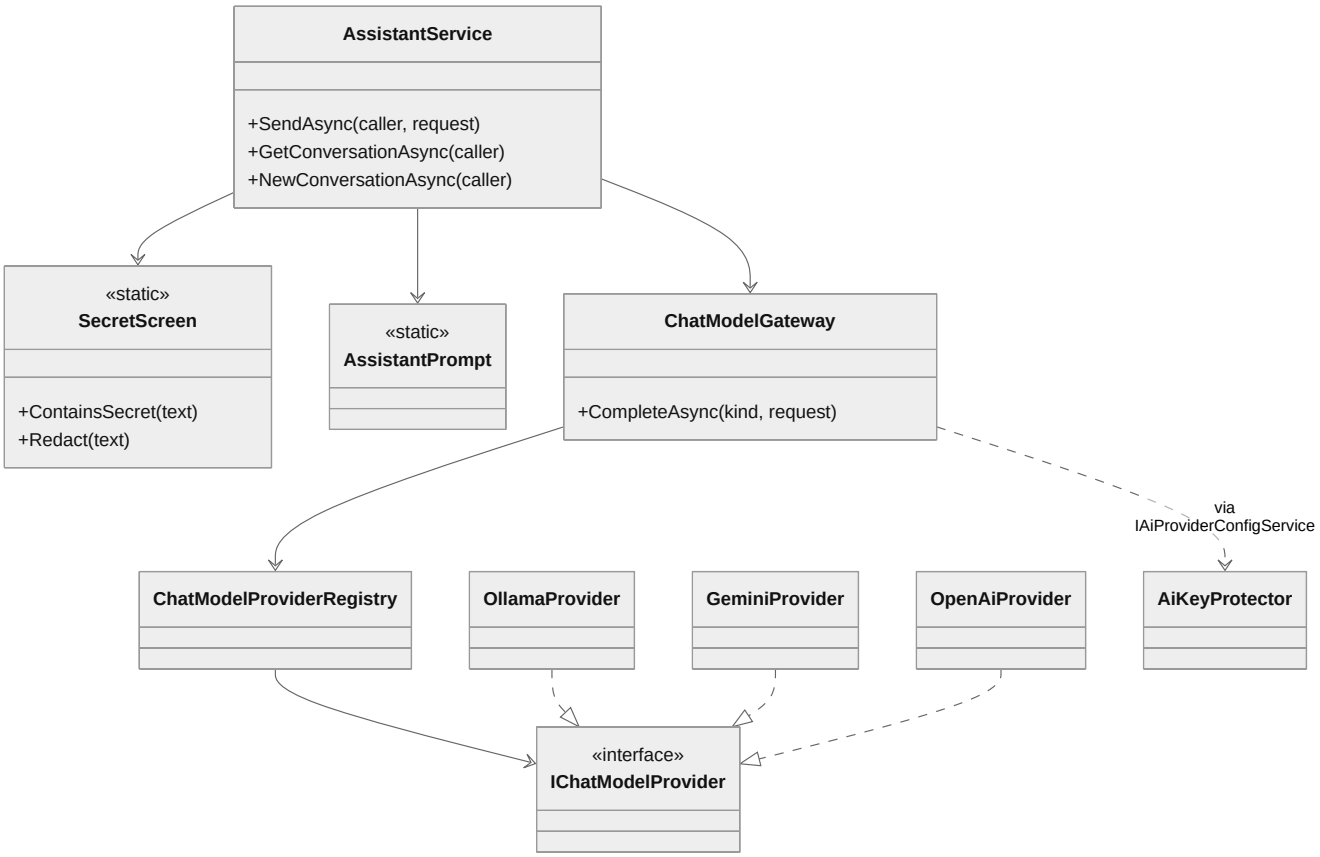
Layer	Type	File	Responsibility
Controller	`AssistantController.GetConversation`, `SendMessage`, `NewConversation`, `GetProviders`	`Controllers/Assistant/AssistantController.cs`	HTTP routing
Service	`IAssistantService` / `AssistantService` (`GetConversationAsync`, `SendAsync`, `NewConversationAsync`; private `LoadHistoryAsync`, `RunTurnAsync`, `ToDtosAsync`)	`Services/Assistant/Chat/`	Conversation engine and tool loop
Prompt / texts	`AssistantPrompt.Build`, `AssistantTexts` (`SecretBlocked`, `EmptyAnswer`, `ActionPrepared`, `PasswordGuidance`, `AdminGuidance`, `PermissionDenied`), `ProviderFailureMessages` (`For`, `NotConfigured`)	`Services/Assistant/Chat/`	Fixed texts
Gateway	`IChatModelGateway` / `ChatModelGateway.CompleteAsync(kind, request)`	`Services/Assistant/ModelGateway/`	Runtime config + provider dispatch
Providers	`IChatModelProvider`; `OllamaProvider`, `GeminiProvider`, `OpenAiProvider`; `ChatModelTypes` (`ChatModelRequest`, `ChatModelResult`, `ChatToolDefinition`, `ChatToolCall`), `AiProviderRuntime`, `ProviderFailure`, `ProviderHttp`	`Integrations/Ai/Providers/`	HTTP to the AI provider, failure classification
Registry	`IChatModelProviderRegistry` / `ChatModelProviderRegistry.Get(kind)`	`Integrations/Ai/Registry/`	Provider by `AiProviderKind`
Key protection	`IAiKeyProtector` / `AiKeyProtector`; `KeyRingProtectionCheck`	`Integrations/Ai/KeyProtection/`	Decrypt API key for one call; startup check

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Layer	Type	File	Responsibility
Rules	`AssistantRules` (`MaxMessageLength` 2000, `HistoryMessages` 12, `HistoryCharacters` 6000, `MaxToolRounds` 4, `ConversationMessages` 100); `SecretScreen` (`ContainsSecret`, `Redact`, `RedactedText = "[hidden]"`)	`Services/Assistant/Rules/`	Limits and secret screening
Repositories	`IAssistantConversationRepository` / `AssistantConversationRepository`; `IAssistantMessageRepository` / `AssistantMessageRepository`; `IAssistantDirectoryRepository.GetFullNameAsync`	`Repositories/Assistant/Conversations/`, `Messages/`, `People/`	Persistence
DTOs	`SendAssistantMessageRequest(Text, Provider)`, `AssistantConversationDto`, `AssistantReplyDto`, `AssistantMessageDto`, `AssistantLinkDto`, `AssistantMessageMeta`, `PrioritisedTask`, `AssistantFailureDto`; `AiProviderOptionDto` (Settings)	`Dtos/Assistant/`; `Dtos/Settings/TfaSettings/AiProviderDtos.cs`	Contracts
Entities	`AssistantConversation`, `AssistantMessage`	`Models/Entities/`	Tables
Frontend	`TfaPanelComponent`, `TfaHeaderComponent`, `TfaComposerComponent` (`TFA_SUGGESTIONS`), `TfaMessageComponent`	`modules/assistant/components/`	Panel, model selector, composer, messages
Frontend service	`AssistantService` (`open`, `close`, `toggle`, `loadProviders`, `loadConversation`, `selectProvider`, `isAcknowledged`, `send`, `acknowledgeCloud`, `retry`, `newChat`); `TFA_PROVIDER_KEY`, `TFA_CLOUD_ACK_KEY`, `PROVIDER_REFRESH_MS`	`modules/assistant/services/assistant.service.ts`	Signals, HTTP, browser settings

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Class diagram



API contracts

#	Method	Route	Permission / policy	Request	Response	Status codes
1	GET	`/api/v1/assistant/conversation`	`[Authorize]` + `perm:assistant.chat`	–	`AssistantConversationDto(ConversationId, Messages)` (newest 100)	200, 401, 403
2	POST	`/api/v1/assistant/conversation/messages`	`[Authorize]` + `perm:assistant.chat`	`SendAssistantMessageRequest(Text, Provider)`	`AssistantReplyDto(Messages, Failure)`	200 (incl. provider failure), 400, 401, 403, 429
3	POST	`/api/v1/assistant/conversation/new`	`[Authorize]` + `perm:assistant.chat`	–	–	204, 401, 403
4	GET	`/api/v1/assistant/providers`	`[Authorize]` + `perm:assistant.chat`	–	`IReadOnlyList<AiProviderOptionDto>` (kind, displayName, modelName, locality; no key data)	200, 401, 403

Request / response example

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```
// POST /api/v1/assistant/conversation/messages
{ "text": "What should I prioritise?", "provider": "Gemini" }
```

```
// 200 OK – answered
{ "messages": [
    { "id": 41, "role": "User", "content": "What should I prioritise?", "kind": "Text",
"provider": null,
      "links": [], "rankedTasks": [], "action": null },
    { "id": 42, "role": "Assistant", "content": "Start with TSK-12, it is 2 days overdue.",
"kind": "Text",
      "provider": "Gemini", "links": [ { "label": "TSK-12", "route": "/tasks/12" } ],
      "rankedTasks": [ { "rank": 1, "taskId": 12, "taskCode": "TSK-12", "title": "Fix login",
"priority": "High",
        "status": "InProgress", "deadline": "2026-10-29T12:30:00Z", "dueDate": "2026-10-29",
"daysUntilDue": -2,
        "stage": "Overdue", "isBlocked": false } ], "action": null } ],
  "failure": null }
```

```
// 200 OK – provider failure (nothing stored)
{ "messages": [],
  "failure": { "kind": "RateLimited", "provider": "Gemini", "retryAfterSeconds": 30,
    "message": "Google Gemini's free limit or rate limit has been reached. Try again in about
30 seconds, or choose another model.",
    "alternatives": [ { "kind": "Ollama", "displayName": "Ollama", "modelName": "llama3.2:3b",
"locality": "Local" } ] } }
```

Service logic – `SendAsync(caller, request)`

1. Trim `Text`; empty or longer than `AssistantRules.MaxMessageLength` (2000) → `AssistantRequestException("A message must be between 1 and 2000 characters.")` (400).
2. `SecretScreen.ContainsSecret(text)` (password phrase with a digit/symbol value, `sk-…`, `AIza…`, `Bearer …`, JWT, 32+ character mixed token) → return one transient `Notice` message (`AssistantTexts.SecretBlocked`, id 0). Nothing is sent or stored.
3. `IAiProviderConfigService.GetAvailableAsync` → use `request.Provider` or the first available; none → `AssistantRequestException("TFA is not available yet. An administrator needs to configure an AI provider.")`; chosen one missing → `ProviderFailureMessages.NotConfigured(kind)` (400).
4. `IAssistantRateLimiter.TryAcquire(userId)`; a wait value → `AssistantRateLimitedException` (429, `Retry-After`) – 4.1.4.
5. `FindLatestForUpdateAsync(userId)` or create a new `AssistantConversation` (latest by `Id` is the active one).
6. `LoadHistoryAsync`: newest 24 (`HistoryMessages * 2`) messages, drop `Notice` metas, take 12, stop at 6000 characters, reverse to chronological.
7. `AssistantPrompt.Build(userName, caller.Role, IAppClock.TodayLocalAsync, application name)`; `userName` from `IAssistantDirectoryRepository.GetFullNameAsync`.
8. `RunTurnAsync`: up to `MaxToolRounds` (4) rounds; tools from `IAssistantToolbox.DefinitionsForAsync(caller)`, none in the last round. Each tool call → `IAssistantToolbox.ExecuteAsync` (4.1.2); collect `Links`, `RankedTasks`, `PendingActionId`. A `ToolOutcome.FixedReply` ends the turn with that exact text. Tool results pass

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`SecretScreen.Redact` before going back to the model. Empty model text → `AssistantTexts.EmptyAnswer`.

9. `ProviderFailureException` → `AssistantReplyDto([], AssistantFailureDto(kind, provider, message, retryAfterSeconds, alternatives))`; nothing stored.
10. Success → insert user and assistant `AssistantMessage` rows (assistant `CreatedAt` = now + 1 ms, `ProviderKind`, `MetaJson` = `AssistantMessageMeta`), set `conversation.UpdatedAt`, `SaveChangesAsync`, map with `ToDtosAsync` (cards from `IAssistantActionCards.GetCardsAsync`).

Service logic – `GetConversationAsync(caller)`: latest conversation of `UserId`; newest `ConversationMessages` (100) messages reversed; none → `AssistantConversationDto(null, [])`.

Service logic – `NewConversationAsync(caller)`: insert a new `AssistantConversation`; it becomes the latest.

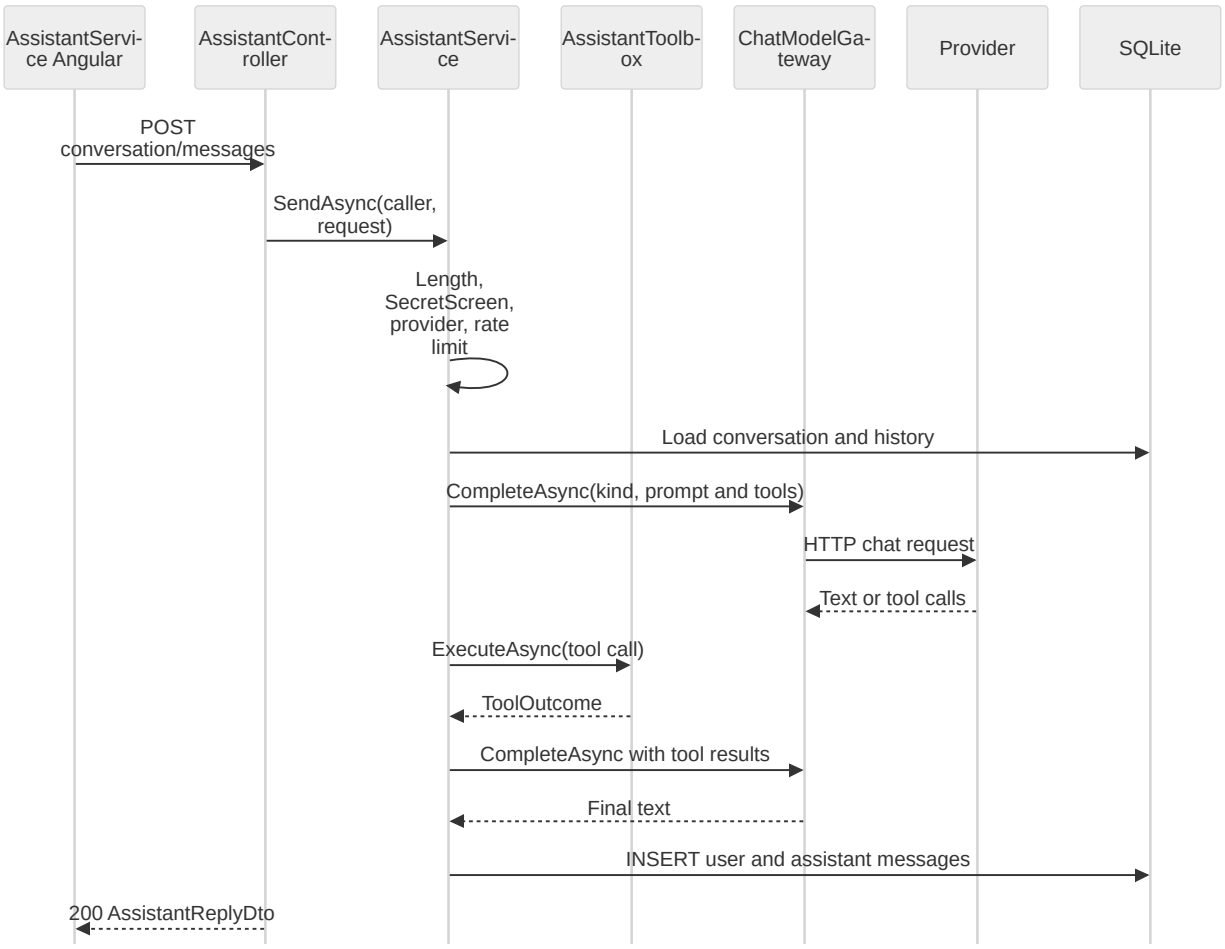
Gateway and providers: `ChatModelGateway.CompleteAsync(kind, request)` → `IAiProviderConfigService.GetRuntimeAsync(kind)` (decrypts the key with `IAiKeyProtector`; `null` → `ProviderNotConfiguredException`) → `IChatModelProviderRegistry.Get(kind).CompleteAsync(runtime, request with ModelName)`.

Provider	Endpoint	Auth	Locality
`OllamaProvider`	`{BaseUrl or http://localhost:11434}/api/chat`	none	Local
`GeminiProvider`	`https://generativelanguage.googleapis.com/v1beta/models/{model}:generateContent`	`x-goog-api-key` header	Cloud
`OpenAiProvider`	`https://api.openai.com/v1/chat/completions`	Bearer header	Cloud

Failure classification (`ProviderFailureKind`): 429 / `insufficient_quota` / `rate_limit_exceeded` / `RESOURCE_EXHAUSTED` → `RateLimited`; 401, 403 / `invalid_api_key` / `UNAUTHENTICATED` / `PERMISSION_DENIED` → `InvalidKey`; 404 / `model_not_found` / `NOT_FOUND` → `ModelUnavailable`; `HttpClient` timeout → `Timeout`; `HttpRequestException` → `Unreachable`; unparsable or empty reply → `Unexpected`. Ollama retries once without tools when the model does not support tools.

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Sequence diagram



Repository methods

Method	Query / purpose	Notes
`IAssistantConversationRepository.FindLatestForUpdateAsync(userId)`	Active conversation	Tracked; `Where UserId`, `OrderByDescending Id`, first
`IAssistantMessageRepository.GetNewestAsync(conversationId, take)`	Newest messages, newest first	`AsNoTracking`, `OrderByDescending Id`, `Take`
`IAssistantDirectoryRepository.GetFullNameAsync(userId)`	Name for the system prompt	`AsNoTracking`, projection
`Add(entity)` (`IBaseRepository`)	Insert conversation or message	`IUnitOfWork.SaveChangesAsync`

Validation rules

Field	Rule	Message
`Text`	1–2000 characters after trim	"A message must be between 1 and 2000 characters."
`Text`	No password or API key/token (`SecretScreen.ContainsSecret`)	"For your safety I did not send or keep that message: it looks like it contains a password or an API key. Please never type passwords into chat. To change your password, open My Profile and use the Change password form."

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Field	Rule	Message
`Provider`	Configured and enabled provider, or omitted	"TFA is not available yet. An administrator needs to configure an AI provider." / "`{Provider}` is not configured. Choose another model, or ask an administrator to set it up."

Error cases

Case	Exception	Status
Length outside 1–2000	`AssistantRequestException`	400
No / unknown provider	`AssistantRequestException`	400
Secret detected	– (Notice message)	200
Provider failure	– (`failure` object, `ProviderFailureMessages.For`)	200
Over 20 per minute	`AssistantRateLimitedException`	429
Client fallbacks	`apiErrorMessage`	"TFA could not answer. Please try again." / "Could not start a new chat."

State machine: *Not applicable for this child module.*

4.1.2 Child module: Actions and Tools

Class / file inventory

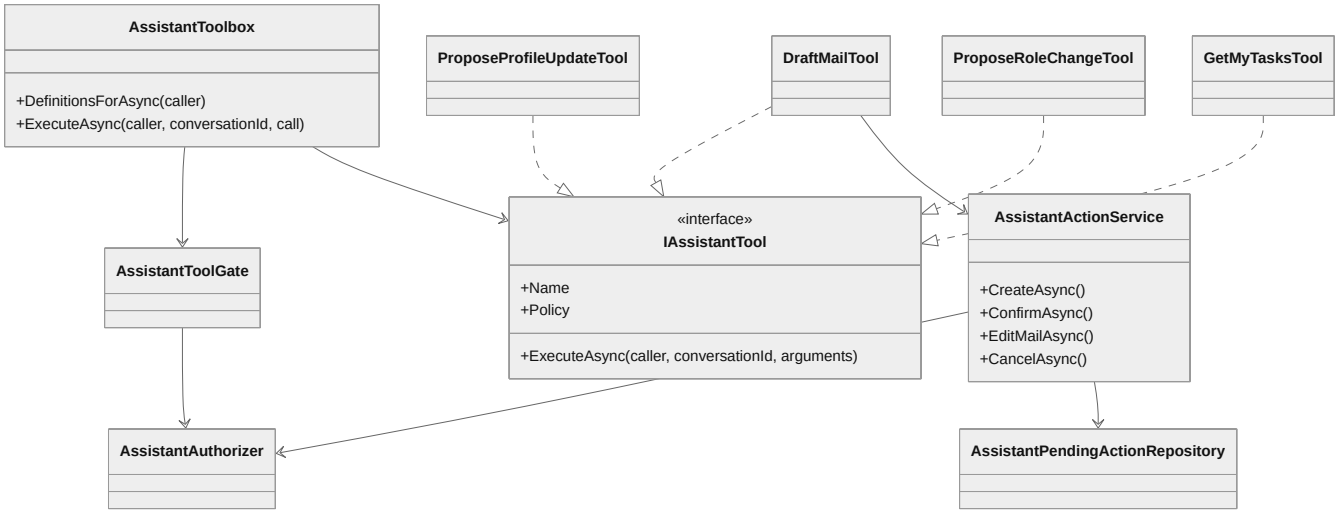
Layer	Type	File	Responsibility
Controller	`AssistantController.EditAction`, `ConfirmAction`, `CancelAction`; `RunActionAsync` (null → 404)	`Controllers/Assistant/AssistantController.cs`	HTTP routing
Service	`IAssistantActionService` / `AssistantActionService` (`CreateAsync`, `EditMailAsync`, `ConfirmAsync`, `CancelAsync`, `GetCardsAsync`; private `ExecuteAsync`, `EnsurePending`, `ResolveRecipientsAsync`, `ToCard`)	`Services/Assistant/Actions/`	Pending-action lifecycle
Toolbox	`IAssistantToolbox` / `AssistantToolbox` (`DefinitionsForAsync`, `ExecuteAsync`), `AssistantToolGate`, `ToolOutcome`	`Services/Assistant/Toolbox/`	Advertise and gate tools
Tools (read)	`GetMyTasksTool`, `PrioritiseTasksTool`, `SearchHelpTool`, `FindUserTool`, `ExplainPasswordChangeTool`, `ContactAdminGuidanceTool`; helper `AssistantTaskReader`	`Services/Assistant/Tools/AssistantReadTools.cs`	Lookups and guidance

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Layer	Type	File	Responsibility
Tools (action)	`DraftMailTool`, `ProposeProfileUpdateTool`, `ProposeRoleChangeTool`; helpers `AssistantUserResolver`, `AssistantManagerResolver`	`Services/Assistant/Tools/AssistantActionTools.cs`	Prepare pending actions
Rules	`AssistantActionRules` (`Lifetime` 10 min, `Requirement(kind)`), `TaskPrioritiser.Rank`	`Services/Assistant/Rules/`	Expiry, confirm policy, ranking
Authorization	`IAssistantAuthorizer` / `AssistantAuthorizer.IsAllowedAsync(principal, policyName)` (`IAuthorizationService.AuthorizeAsync`)	`Authorization/Assistant/`	Same `perm:{key}` policies as REST
Repositories	`IAssistantPendingActionRepository` / `AssistantPendingActionRepository`; `IAssistantDirectoryRepository` / `AssistantDirectoryRepository`	`Repositories/Assistant/PendingActions/`, `People/`	Actions, people, roles
DTOs	`SendMailPayload`, `UpdateProfilePayload`, `ChangeRolePayload`, `MailRecipientRef`, `EditMailActionRequest`, `AssistantActionResultDto`, `ResolvedUser`; `AssistantPendingActionDto`, `AssistantActionFieldDto`	`Dtos/Assistant/AssistantActionDtos.cs`, `AssistantDtos.cs`	Contracts
Validators (reused)	`SendMailRequestValidator` (`MailRequestValidators.cs`), `UpdateOwnProfileRequestValidator`	`Dtos/Communication/Mail/Validators/`, `Dtos/Profile/Validators/`	Mail and profile rules
Other services	`ITaskService.ListAsync`, `GetDetailsAsync`; `IMailService.SendAsync`, `GetContactsAsync`; `IProfileService.UpdateAsync`; `IEmployeeService.GetByIdAsync`, `UpdateAsync`	Task, Mail, Profile, Employee modules	Reads and executions
Frontend	`TfaActionCardComponent` (Confirm / Edit / Cancel, edit form, status labels); `AssistantService.confirm`, `cancel`, `edit`	`modules/assistant/components/`, `services/`	Action Card

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Class diagram



API contracts

#	Method	Route	Permission / policy	Request	Response	Status codes
5	PUT	`/api/v1/assistant/actions/{id}` (`{id:guid}`)	`[Authorize]` + `perm:assistant.actions.confirm`	`EditMailActionRequest(To, Subject, Body)`	`AssistantPendingActionDto`	200, 400, 401, 403, 404, 409
6	POST	`/api/v1/assistant/actions/{id}/confirm`	`[Authorize]` + `perm:assistant.actions.confirm`; `perm:roles.manage` re-checked in the service for `ChangeUserRole`	–	`AssistantActionResultDto(Id, Status, Message, Reply)`	200 (incl. `Refused`, `Failed`), 401, 403, 404, 409
7	POST	`/api/v1/assistant/actions/{id}/cancel`	`[Authorize]` + `perm:assistant.actions.confirm`	–	`AssistantPendingActionDto`	200, 401, 403, 404

Model tools (not HTTP; called by the model inside `POST /api/v1/assistant/conversation/messages`)

Tool name	Class	Policy	Arguments	Effect
`get_my_tasks`	`GetMyTasksTool`	none	`filter` (upcoming, due_today, overdue, all_open), `scope` (mine, accessible), `limit` (default 10, max 20)	`AssistantTaskReader` → `ITaskService.ListAsync(…, userId, role)`; links `/tasks/{id}`
`prioritise_tasks`	`PrioritiseTasksTool`	none	`limit` (default 5, max 10)	`TaskPrioritiser.Rank` on own open tasks → `RankedTasks`
`search_help`	`SearchHelpTool`	none	`topic`	`IHelpKnowledgeBase.Search(topic, role)` (4.1.3)

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Tool name	Class	Policy	Arguments	Effect
`find_user`	`FindUserTool`	none	`name`	`IMailService.GetContactsAsync` (max 10, active users)
`explain_password_change`	`ExplainPasswordChangeTool`	none	–	Fixed `AssistantTexts.PasswordGuidance` + `/profile` link
`contact_admin_guidance`	`ContactAdminGuidanceTool`	none	–	Fixed `AssistantTexts.AdminGuidance`
`draft_mail`	`DraftMailTool`	none	`recipient` ("manager"), `to[]`, `taskId`, `subject`, `body`	`SendMail` pending action
`propose_profile_update`	`ProposeProfileUpdateTool`	none	`fullName`, `email`	`UpdateOwnProfile` pending action
`propose_role_change`	`ProposeRoleChangeTool`	`perm:roles.manage` (`AssistantPolicies.RolesManage`)	`user`, `role`	`ChangeUserRole` pending action

No tool takes a user id, role or password; the caller is always `AssistantCaller`.

Request / response example

```
// PUT /api/v1/assistant/actions/3f2b0c1e-9a7d-4c55-8f0e-1b2c3d4e5f60
{ "to": ["ravi.kumar@taskflow.com"], "subject": "TSK-12 update", "body": "Hi Ravi, ..." }
```

```
// 200 OK (AssistantPendingActionDto)
{ "id": "3f2b0c1e-9a7d-4c55-8f0e-1b2c3d4e5f60", "kind": "SendMail", "status": "Pending",
  "expiresAt": "2026-10-31T10:10:00Z", "title": "Mail draft",
  "fields": [ { "label": "To", "value": "Ravi Kumar <ravi.kumar@taskflow.com>" }, { "label":
"Subject", "value": "TSK-12 update" } ],
  "body": "Hi Ravi, ...", "canEdit": true }
```

```
// POST /api/v1/assistant/actions/3f2b0c1e-9a7d-4c55-8f0e-1b2c3d4e5f60/confirm  ->  200 OK
{ "id": "3f2b0c1e-9a7d-4c55-8f0e-1b2c3d4e5f60", "status": "Confirmed",
  "message": "Done. Your mail \"TSK-12 update\" was sent to Ravi Kumar.",
  "reply": { "id": 57, "role": "Assistant", "content": "Done. Your mail \"TSK-12 update\" was
sent to Ravi Kumar.", "kind": "Text" } }
```

```
// 409 Conflict
{ "status": 409, "title": "This action can no longer be done", "detail": "This request has
expired. Please ask TFA again." }
```

Service logic – `CreateAsync(caller, conversationId, kind, payload)`

1. New `AssistantPendingAction`: `Id = Guid`, `Status = Pending`, `CreatedAt = now`, `ExpiresAt = now + AssistantActionRules.Lifetime` (10 min), `PayloadJson` = serialised `SendMailPayload` /

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`UpdateProfilePayload` / `ChangeRolePayload`.
2. `SaveChangesAsync`; audit `prepared` (4.1.4); return `ToCard` (titles "Mail draft", "Update your profile", "Change role").
3. The tool returns `ToolOutcome` with `FixedReply = AssistantTexts.ActionPrepared` and `PendingActionId`.

Service logic – `ConfirmAsync(caller, id)`

1. `FindForUserAsync(id, userId)`; `null` → controller 404.
2. Pending and `ExpiresAt <= now` → `SetStatusAsync(Expired)`, audit `expired`, `AssistantActionConflictException("This request has expired. Please ask TFA again.")`.
3. Not pending → 409: `Confirmed` → "This action has already been done."; else "This action can no longer be confirmed (`{status}`). Please ask TFA again."
4. `AssistantActionRules.Requirement(kind)` → `IAssistantAuthorizer.IsAllowedAsync(caller.Principal, policy)`; refused → `Refused`, audit `refused`, store a `Refusal` message with `AssistantTexts.PermissionDenied(description, who)`, return 200 with `Status = Refused`.
5. `TryClaimAsync(id, now)` (single `ExecuteUpdateAsync` Pending → Confirmed); false → 409 "This action has already been done."
6. `ExecuteAsync`:
   - `SendMail`: re-validate with `SendMailRequestValidator`; `IMailService.SendAsync(userId, request)` (no Cc, no attachments).
   - `UpdateOwnProfile`: re-validate with `UpdateOwnProfileRequestValidator`; `IProfileService.UpdateAsync(userId, new UpdateOwnProfileRequest(FullName, Email))` – mobile `null`, kept.
   - `ChangeUserRole`: `IEmployeeService.GetByIdAsync`, then `IEmployeeService.UpdateAsync` with all existing fields and the new `RoleId` (the employee service rotates the target's `SecurityStamp` and revokes refresh tokens).
7. Exception (not cancellation) → `Failed`, audit `failed`, reply "I could not do that: `{reason}`. Nothing was sent." / "... Nothing was changed." (`reason` = message of `AppException` / `ValidationException`, except concurrency/unique errors; otherwise "something went wrong on the server", logged).
8. Success → audit `confirmed` (an audit failure is logged and never turns success into failure); store the reply message; return `AssistantActionResultDto`.

Service logic – `EditMailAsync(caller, id, request)`: `FindForUserForUpdateAsync`; `EnsurePending` (409 "This action can no longer be changed. Please ask TFA again."); kind must be `SendMail` (else 400 "Only a mail can be edited."); `SendMailRequestValidator`; `ResolveRecipientsAsync` (`GetActiveRecipientsAsync` on lower-case addresses; unknown → 400 "These are not active TaskFlow users: `{addresses}`."); replace `PayloadJson`.

Service logic – `CancelAsync(caller, id)`: tracked lookup; still pending and not expired → `Cancelled`, `ResolvedAt`, audit `cancelled`; always returns the current card.

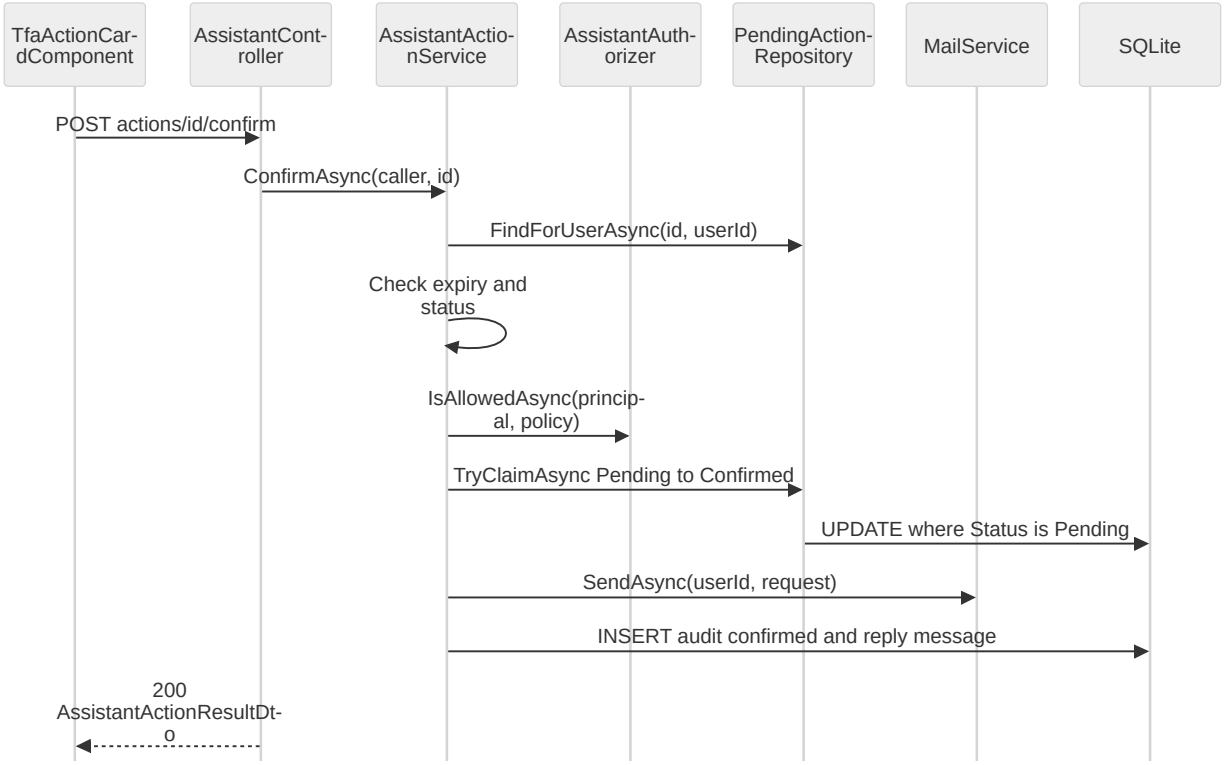
Tool-level logic

- `AssistantToolbox.DefinitionsForAsync`: only tools whose `Policy` passes `AssistantToolGate` are advertised.

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- `AssistantToolbox.ExecuteAsync`: unknown tool → message to the model; gate refused → audit `refused` (when `ActionKind` is set) and fixed `Refusal` reply `AssistantTexts.PermissionDenied`; arguments must be a JSON object.
- `AssistantUserResolver.FindAsync`: `@` → exact active email; else exact full name/username (single match wins); else partial `LIKE` (max 5); ambiguity returns candidates.
- `AssistantManagerResolver.ResolveAsync`: with `taskId`, `ITaskService.GetDetailsAsync(id, userId, role)` (`null` → "That task was not found, or you do not have access to it."); task creator if another active user; else active users with `RoleId == SystemRoles.ManagerId` managing one of the caller's departments – exactly one required.
- `ProposeProfileUpdateTool`: merge with current name/email (`GetNameAndEmailAsync`), validate with `IValidator<UpdateOwnProfileRequest>`, refuse when `EmailUsedByOtherAsync`.
- `ProposeRoleChangeTool`: `FindRoleByNameAsync` (exact), `AssistantUserResolver.FindAsync`, refuse self ("For safety I do not change your own role. Please ask another administrator to do it."), no-op when the role is unchanged.
- `AssistantTaskReader.OpenTasksAsync`: one `ITaskService.ListAsync` query per open status (`ToDo`, `InProgress`, `InQA`, `Blocked`), page size 100, at most 5 pages each.
- `TaskPrioritiser.Rank`: exclude `Completed`, `Cancelled`; order by stage (Overdue 0, DueToday 1, other 2), local due date, priority descending, deadline, task code; `IsBlocked` flag for `Blocked`.

Sequence diagram



Repository methods

Method	Query / purpose	Notes
`IAssistantPendingActionRepository.GetForUserAsync(userId, ids)`	Cards for messages	`AsNoTracking`, `UserId` + `ids.Contains`
`FindForUserAsync(id, userId)`	Confirm lookup	`AsNoTracking`, owner filter

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Method	Query / purpose	Notes
`FindForUserForUpdateAsync(id, userId)`	Edit / cancel lookup	Tracked, owner filter
`TryClaimAsync(id, now)`	Atomic Pending → Confirmed	`ExecuteUpdateAsync` filtered on `Status == Pending`; true when a row changed
`SetStatusAsync(id, status, now)`	Status + `ResolvedAt`	`ExecuteUpdateAsync`
`IAssistantDirectoryRepository.FindActiveAsync`, `FindActiveByEmailAsync`, `FindActiveByExactNameAsync`, `SearchActiveByNameAsync`	Resolve people	Active users only; deleted hidden by the global filter; `LIKE` with `%`, `_`, `\` escaped
`GetNameAndEmailAsync`, `EmailUsedByOtherAsync`	Profile proposal	Case-insensitive email compare
`FindRoleByNameAsync(name)`	Role lookup	`Roles` exact `Name ==`
`GetDepartmentIdsAsync`, `GetActiveManagersOfDepartmentsAsync`	"My manager"	`UserDepartments`; `RoleId == SystemRoles.ManagerId` and `ManagedDepartments`
`GetActiveRecipientsAsync(lowerEmails)`	Mail edit recipients	Active users by lower-case email

Validation rules

Field	Rule	Message
`EditMailActionRequest.To`	1–50 addresses; each an email of an active user	"Add between 1 and 50 recipients." / "'`{value}`' is not an email address." / "These are not active TaskFlow users: `{addresses}`."
`Subject`, `Body`	At least one; subject max 200, body max 20,000 (`SendMailRequestValidator`)	"Write a subject or a message before sending."
Action kind (edit)	`SendMail` only	"Only a mail can be edited."
Action state	Pending and not expired	"This action can no longer be changed. Please ask TFA again."
Profile payload	`UpdateOwnProfileRequestValidator` (name max 200, `@taskflow.com` email max 256)	Reason in "I could not do that: `{reason}`. Nothing was changed."
Role payload	Existing role (exact name), active target, not the requester	"For safety I do not change your own role. Please ask another administrator to do it."

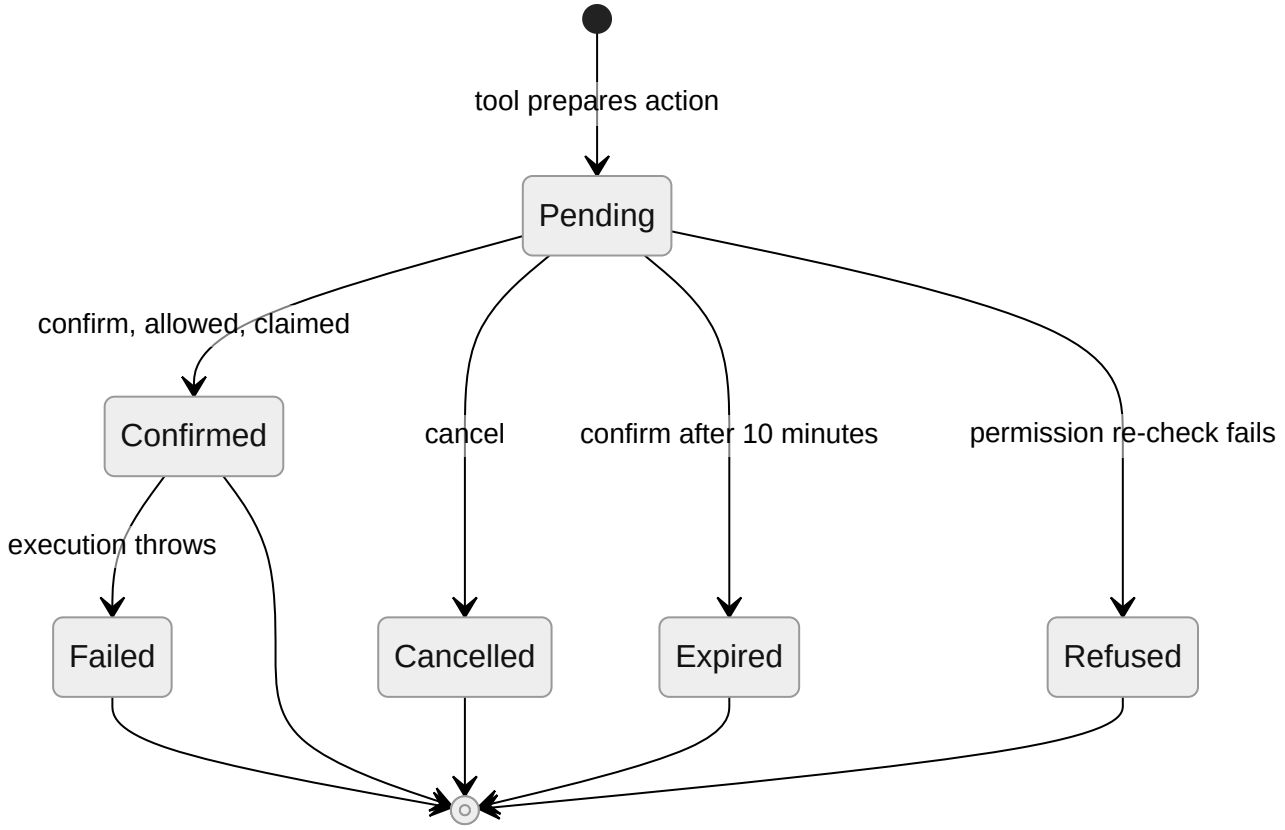
Error cases

Case	Exception	Status
Action unknown / not the caller's	– (`NotFound()`)	404

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Case	Exception	Status
Expired / already done / not pending	`AssistantActionConflictException`	409
Invalid edit	`AssistantRequestException`	400
Permission lost at confirm	– (`Status = Refused`)	200
Execution failure	– (`Status = Failed`)	200
Missing `assistant.actions.confirm`	policy	403
Client fallbacks	`apiErrorMessage`	"Could not confirm that action." / "Could not cancel that action." / "Could not save your changes."

State machine – `AssistantActionStatus`



The claim moves the row to `Confirmed` before execution; a thrown execution then sets `Failed`. `ToCard` shows a pending row past `ExpiresAt` as `Expired` without writing it.

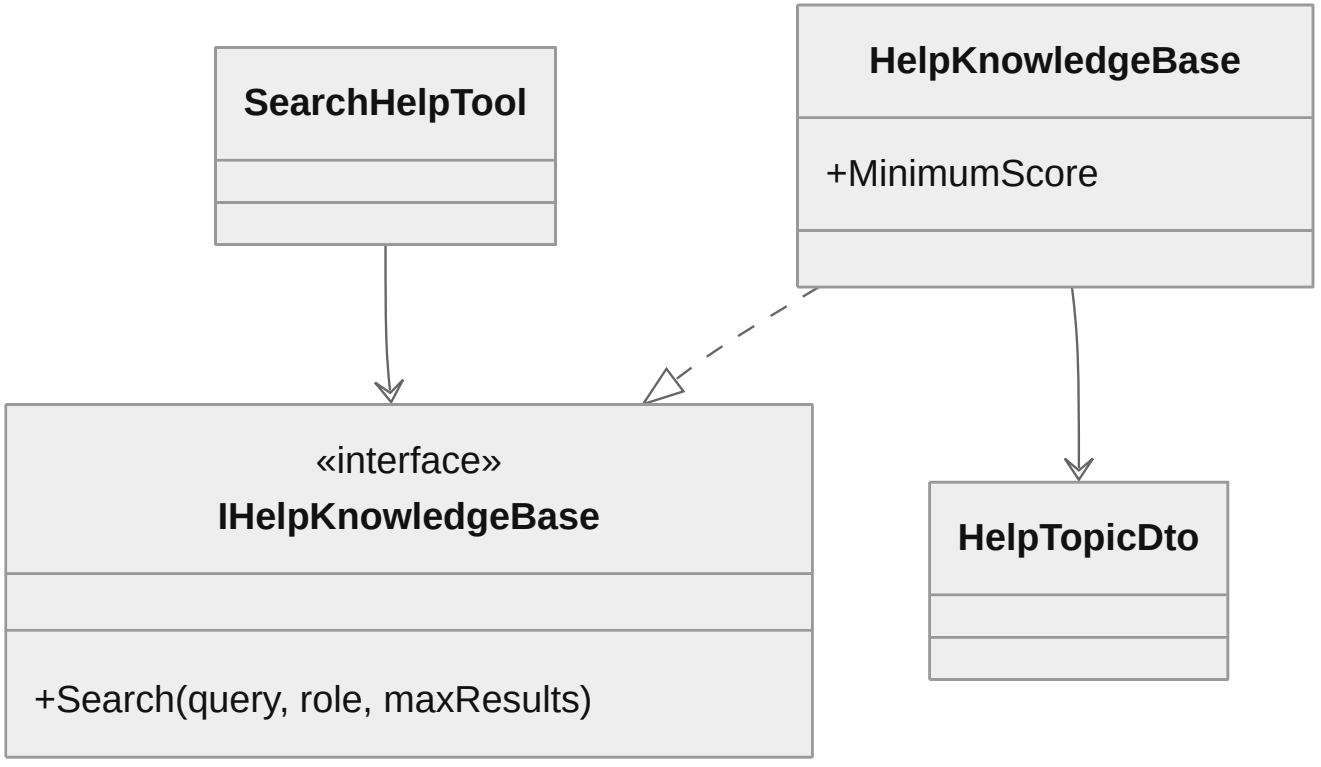
4.1.3 Child module: Help Knowledge

Class / file inventory

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Layer	Type	File	Responsibility
Service	`IHelpKnowledgeBase` / `HelpKnowledgeBase` (`Search(query, role, maxResults = 3)`, `MinimumScore` 3), `HelpTopicDto`	`Services/Assistant/HelpKnowledge/`	Topic search and role redaction
Content	19 topics: `about-tfa.md`, `application-settings.md`, `board.md`, `change-password.md`, `change-role.md`, `change-task-status.md`, `compose-mail.md`, `create-task.md`, `department-management.md`, `edit-profile.md`, `employee-management.md`, `mail-folders.md`, `notifications.md`, `reassign-task.md`, `reminder-settings.md`, `reports-and-workload.md`, `task-comments-and-activity.md`, `task-list-and-filters.md`, `tfa-settings.md`	`Services/Assistant/Knowledge/`	Embedded resources (`<EmbeddedResource Include="Services\Assistant\Knowledge\*.md" />` in `TaskFlow.Api.csproj`)
Caller	`SearchHelpTool`	`Services/Assistant/Tools/AssistantReadTools.cs`	Tool `search_help`
Frontend	`TfaMessageComponent` (link buttons)	`modules/assistant/components/`	Shows answer and screen link

Class diagram



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API contracts: no own endpoint – reached through `search_help` inside `POST /api/v1/assistant/conversation/messages` (`perm:assistant.chat`).

Request / response example (tool result passed to the model, Employee asking how to create a task)

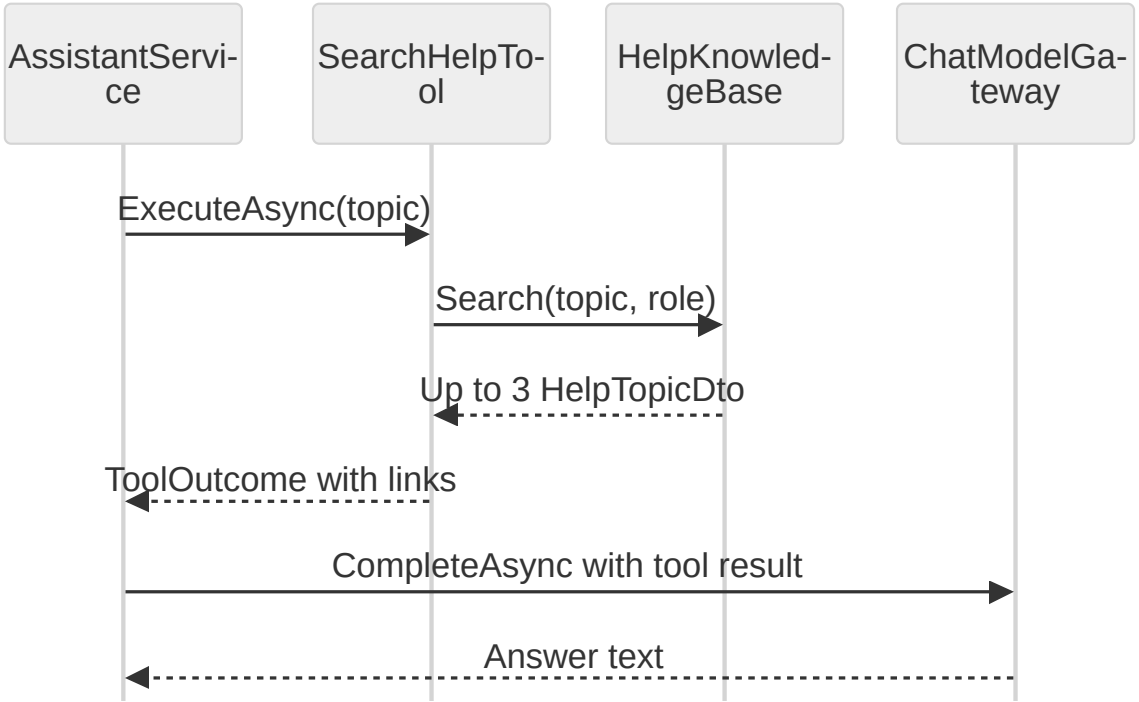
```
{ "found": true, "topics": [ { "title": "Create a task",
  "content": "Tasks are created by Managers and Administrators. If you need a new task, ask
your manager to create and assign it to you.",
  "notAvailableForThisUsersRole": true } ] }
```

Service logic – `Search(query, role, maxResults = 3)`

1. Topics loaded once (singleton) from embedded resources whose name contains `.Assistant.Knowledge.` and ends `.md`; `Topic.Parse` reads the front-matter (`title`, `roles` default `all`, `others`, `keywords`, `link`).
2. Lower-case and tokenise the query (separators, length ≥ 3, stop words removed).
3. Score: multi-word keyword contained in the query +6; single keyword whole-word (plural tolerant) +4; token in keyword words +3; token in title words +2.
4. Keep score ≥ `MinimumScore` (3); order by score, then id; take 3.
5. `ToDto`: `roles` contains `all` or the caller's role name → full body + link; else `others` note (default "`{Title}` is not available to your role."), no link, `IsRestricted = true`.

Role restriction in the shipped files: `Administrator` – `application-settings`, `change-role`, `department-management`, `employee-management`, `tfa-settings`; `Manager, Administrator` – `create-task`, `reassign-task`, `reminder-settings`, `reports-and-workload`; `all` – the other 10.

Sequence diagram



Repository methods: *Not applicable – in-memory list of embedded topics.*

Validation rules: *Not applicable – the chat rules of 4.1.1 apply.*

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Error cases

Case	Exception	Status
No topic scores ≥ 3	– (empty result; the model says it does not have the information)	200
Topic not for the caller's role	– (`IsRestricted` , `others` note)	200

State machine: *Not applicable for this child module.*

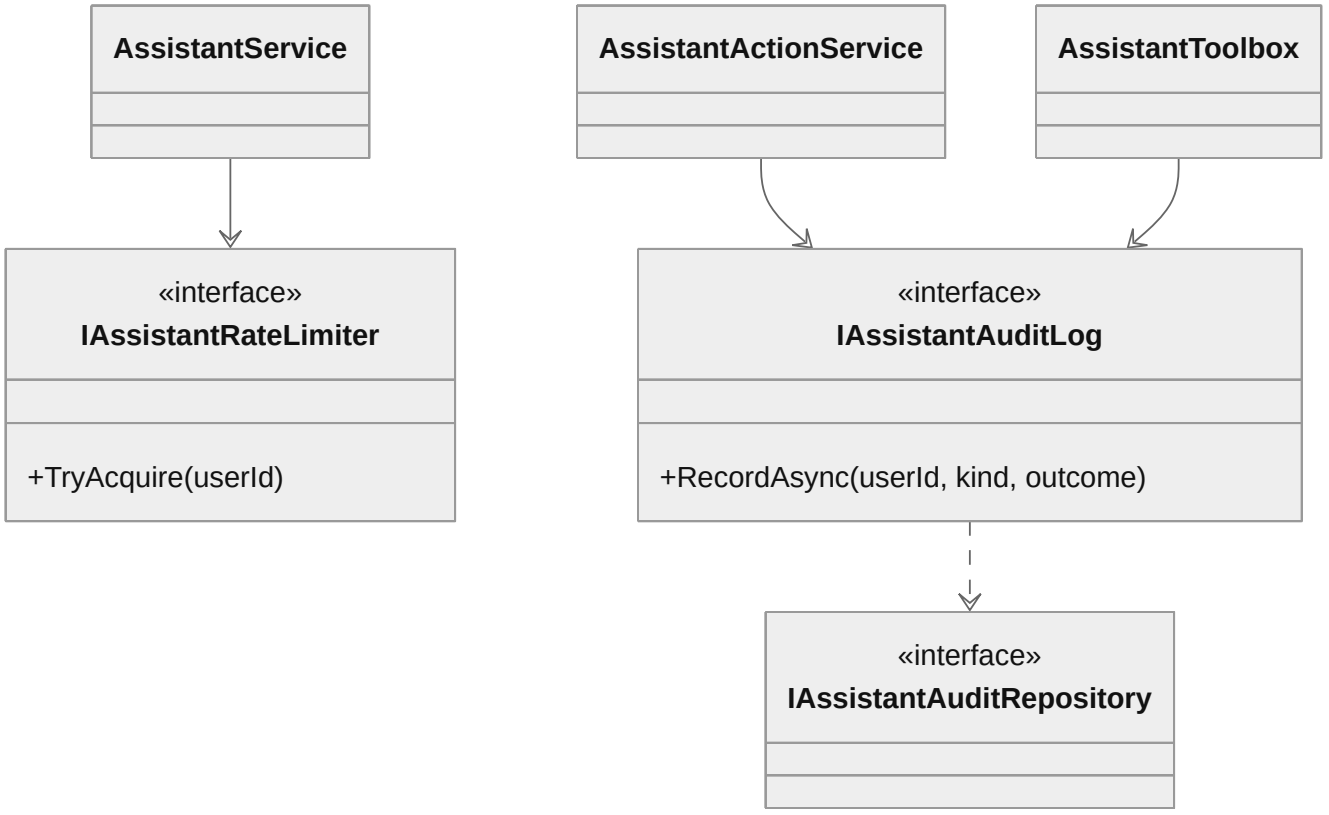
4.1.4 Child module: Audit Log and Rate Limits

Class / file inventory

Layer	Type	File	Responsibility
Service	`IAssistantAuditLog` / `AssistantAuditLog.RecordAsync(userId, kind, outcome)` (scoped)	`Services/Assistant/AuditLog/`	Write `AssistantAudits` rows
Service	`IAssistantRateLimiter` / `AssistantRateLimiter.TryAcquire(userId)` (singleton), `RequestsPerMinute` 20	`Services/Assistant/RateLimits/`	Sliding 1-minute window per user
Repository	`IAssistantAuditRepository` / `AssistantAuditRepository` (`Add` only)	`Repositories/Assistant/Audits/`	Insert
Entity	`AssistantAudit`	`Models/Entities/AssistantAudit.cs`	Table
Exception	`AssistantRateLimitedException`	`Common/Exceptions/AssistantExceptions.cs`	429 + `Retry-After`
Middleware	`GlobalExceptionHandler`	`Middleware/GlobalExceptionHandler.cs`	Problem details and header

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Class diagram



API contracts: no own endpoint – the limiter guards `POST /api/v1/assistant/conversation/messages`; audit rows are written by 4.1.2 and never read by any endpoint.

Request / response example

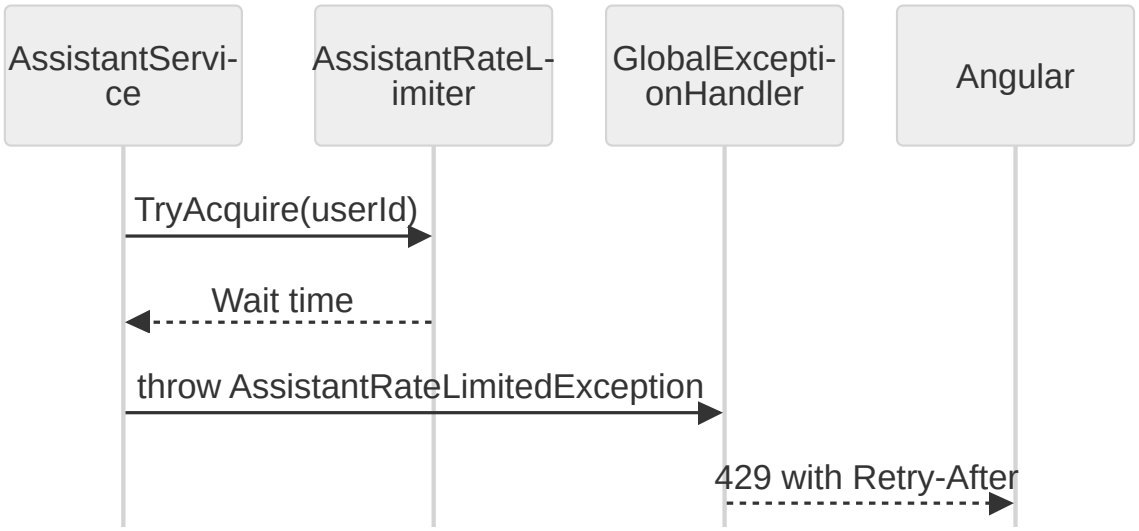
```
// 429 Too Many Requests (header Retry-After: 18)
{ "status": 429, "title": "Too many requests",
  "detail": "You are sending messages too quickly. Please wait about 18 seconds and try again."
}
```

Service logic

- `AssistantRateLimiter.TryAcquire(userId)`: `ConcurrentDictionary<int, Queue<DateTimeOffset>>`; under a per-queue lock drop entries older than 1 minute; count ≥ 20 → return `Window - (now - oldest)`; else enqueue and return `null`. Runs after the length check, `SecretScreen` and provider selection, so only messages passing them are counted.
- `AssistantAuditLog.RecordAsync`: add `AssistantAudit` (`At` = UTC now) and save immediately. Outcomes: `prepared`, `confirmed`, `cancelled`, `expired`, `refused`, `failed`. No message body, password or key is stored.

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Sequence diagram



Repository methods

Method	Query / purpose	Notes
`IAssistantAuditRepository.Add`	Insert audit row	Base repository only; no read methods

Validation rules

Field	Rule	Message
Messages per user	≤ 20 in any 1-minute window	"You are sending messages too quickly. Please wait about `{n}` seconds and try again."

Error cases

Case	Exception	Status
Over the limit	`AssistantRateLimitedException`	429
Audit write fails after a confirmed action	– (logged only)	–

State machine: *Not applicable for this child module.*

4.1.5 Child module: Voice Input

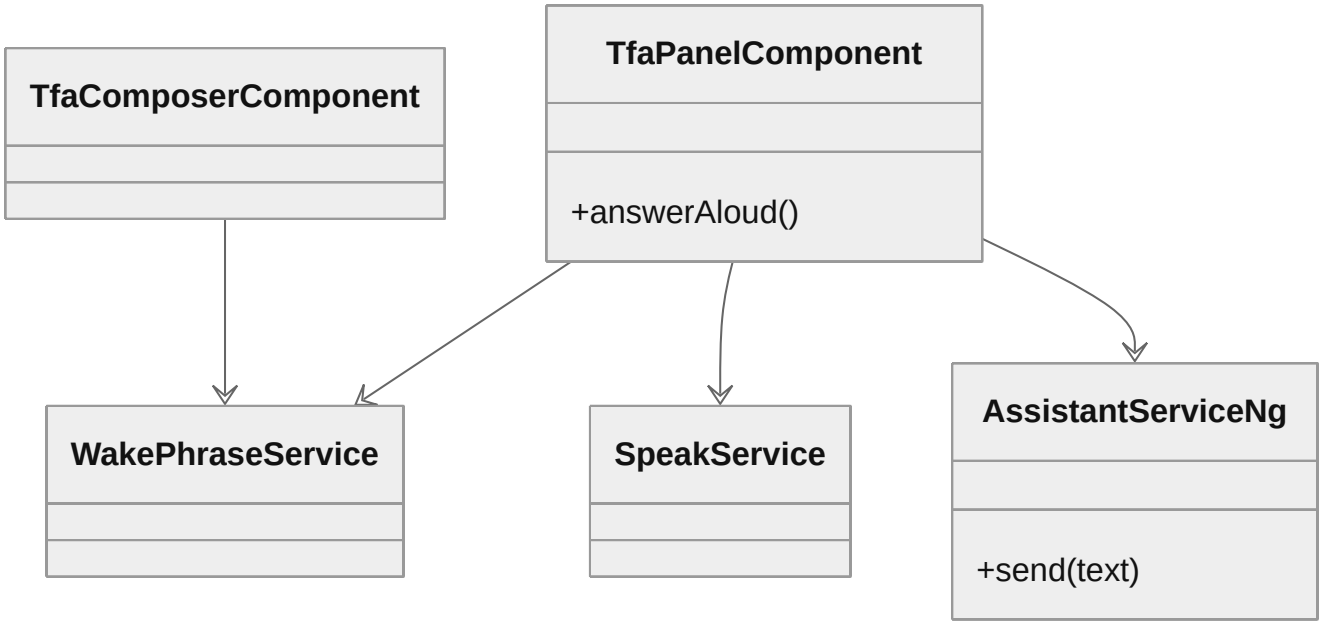
Class / file inventory

Layer	Type	File	Responsibility
Frontend service	`WakePhraseService` (`SPEECH_RECOGNITION_FACTORY`, `TFA_VOICE_KEY = 'tfa.handsfree'`, `COMMAND_SILENCE_MS` 2000, `WAKE_CAPTURE_TIMEOUT_MS` 8000, `DICTATION_TIMEOUT_MS` 10000)	`modules/assistant/services/wake-phrase.service.ts`	Browser `SpeechRecognition`: wake phrase, dictation

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Layer	Type	File	Responsibility
Frontend service	`SpeakService` (`SPEECH_SYNTHESIS_FACTORY`, `UTTERANCE_FACTORY`, `MAX_SPOKEN_CHARS` 500)	`modules/assistant/services/speak.service.ts`	`speechSynthesis`
Models	speech factory interfaces	`services/wake-phrase.model.ts`, `services/speak.model.ts`	Types
Components	`TfaPanelComponent` (`answerAloud`, heard text, voice hint), `TfaComposerComponent` (mic, listening bar, hands-free toggle)	`modules/assistant/components/`	Voice wiring
Backend	–	–	Text goes through `POST /api/v1/assistant/conversation/messages` (4.1.1)

Class diagram



`AssistantServiceNg` stands for the Angular `AssistantService`.

API contracts: no own endpoint.

Request / response example

```
// Spoken "Hey TF assistant, what are my upcoming tasks" is sent as
{ "text": "what are my upcoming tasks", "provider": "Ollama" }
```

Service logic – `WakePhraseService`

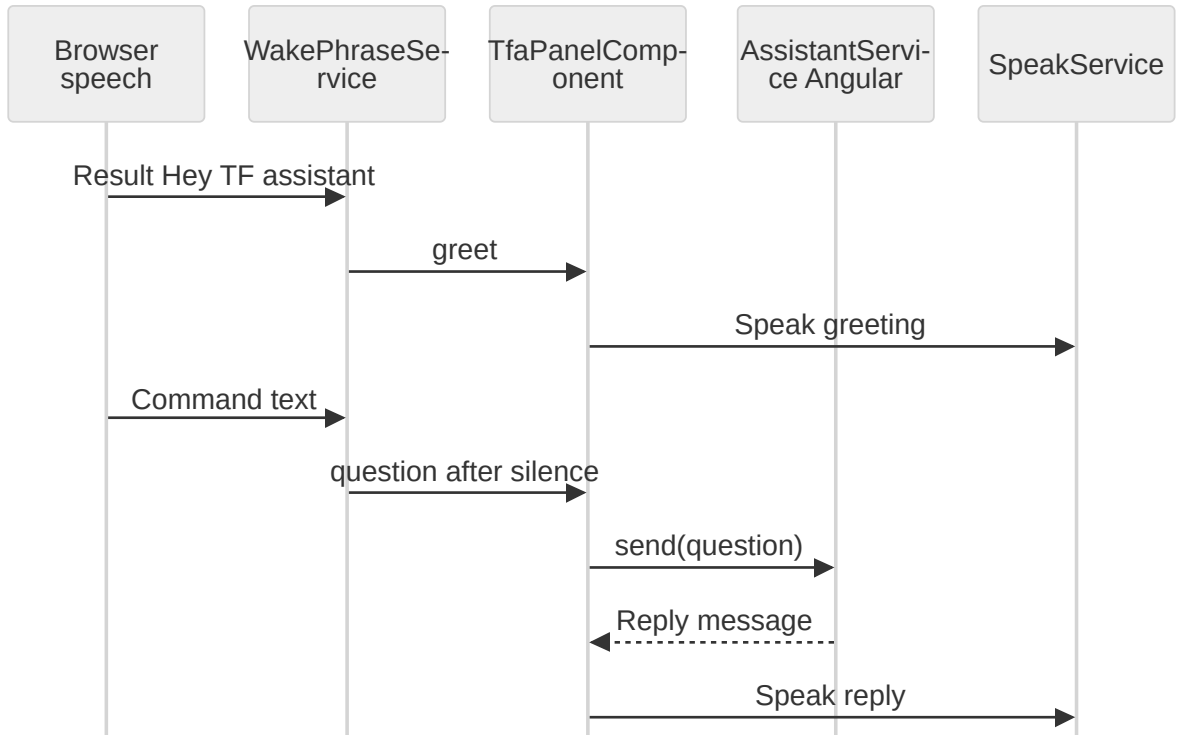
1. The factory creates `SpeechRecognition` / `webkitSpeechRecognition`, or `null` when unsupported.

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2. Hands-free listens continuously for the wake phrase (regex tolerant of "Hey T F assistant", "Hay TFA assistant", "Hey Tifa assistant").
3. Emits `wake`, `greet` (phrase alone), `interim` (live text) and `question` (after 2000 ms of silence); capture timeout 8000 ms; microphone dictation timeout 10000 ms.
4. Restarts ended sessions after 50 ms; quick failures back off from 300 ms and give up after 6 with a reason message.
5. Only a deliberate **Turn off** is stored (`tfa.handsfree`).

Service logic – `SpeakService`: speaks at most 500 characters; `TfaPanelComponent.answerAloud` speaks only replies to spoken questions and pauses recognition while speaking.

Sequence diagram



Repository methods: *Not applicable – browser `localStorage` only.*

Validation rules

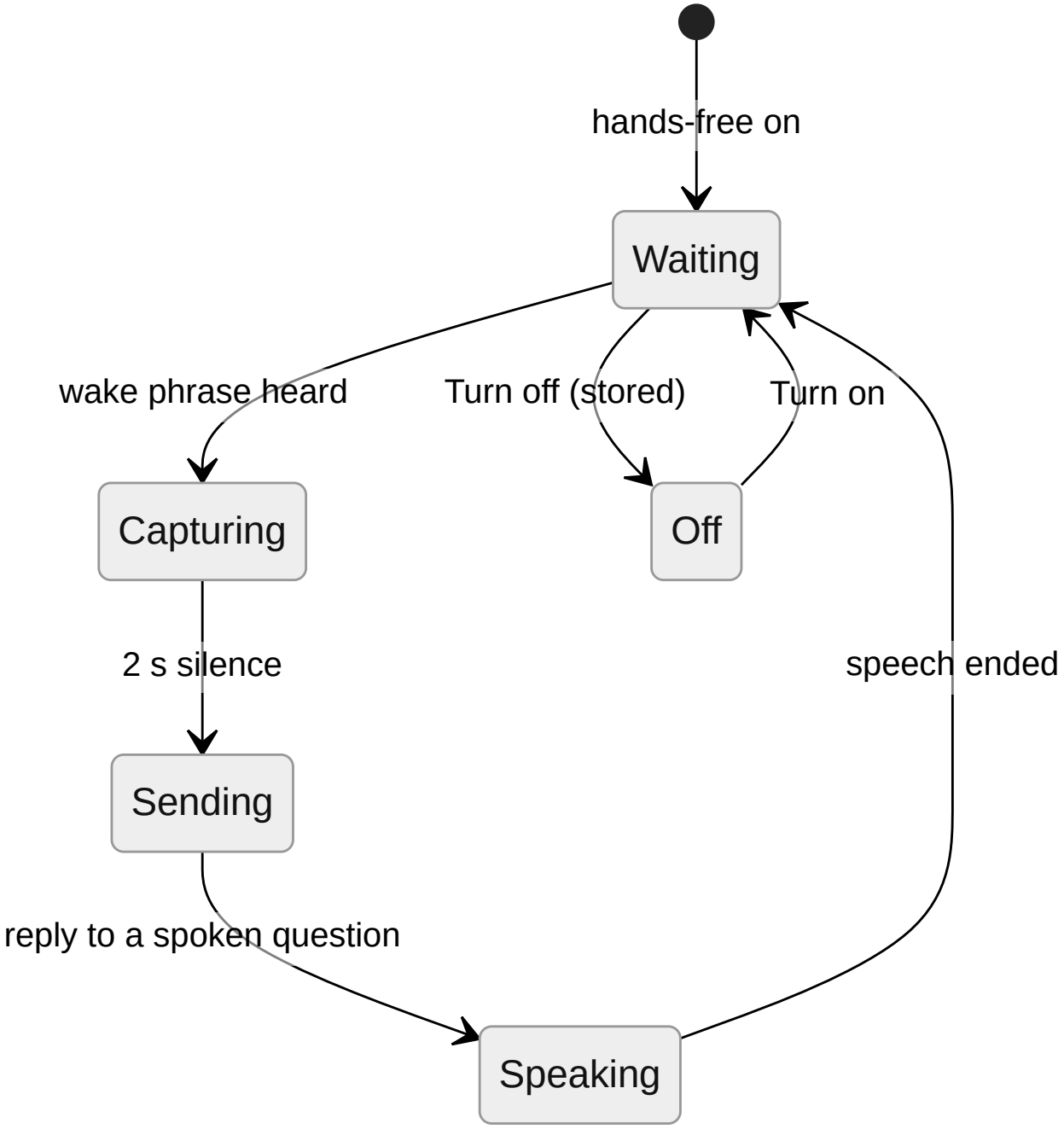
Condition	Rule	Message
No speech recognition	Factory returns `null`	"Voice input is not supported in this browser. Try Chrome or Edge, or type your question."
`not-allowed` / `service-not-allowed`	Microphone blocked	"Microphone access was denied. Allow it in your browser to use voice, or type your question."
`audio-capture`	No microphone	"No microphone was found. You can type your question instead."
`network`	No connection for the speech service	"Voice needs an internet connection in this browser. You can type your question instead."
Auto start blocked	`waitForGesture`	"Click anywhere on the page to turn "Hey TF assistant" back on."

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Condition	Rule	Message
`no-speech` / `aborted`	Ignored	–

Error cases: as the validation table; none reach the server.

State machine – hands-free voice

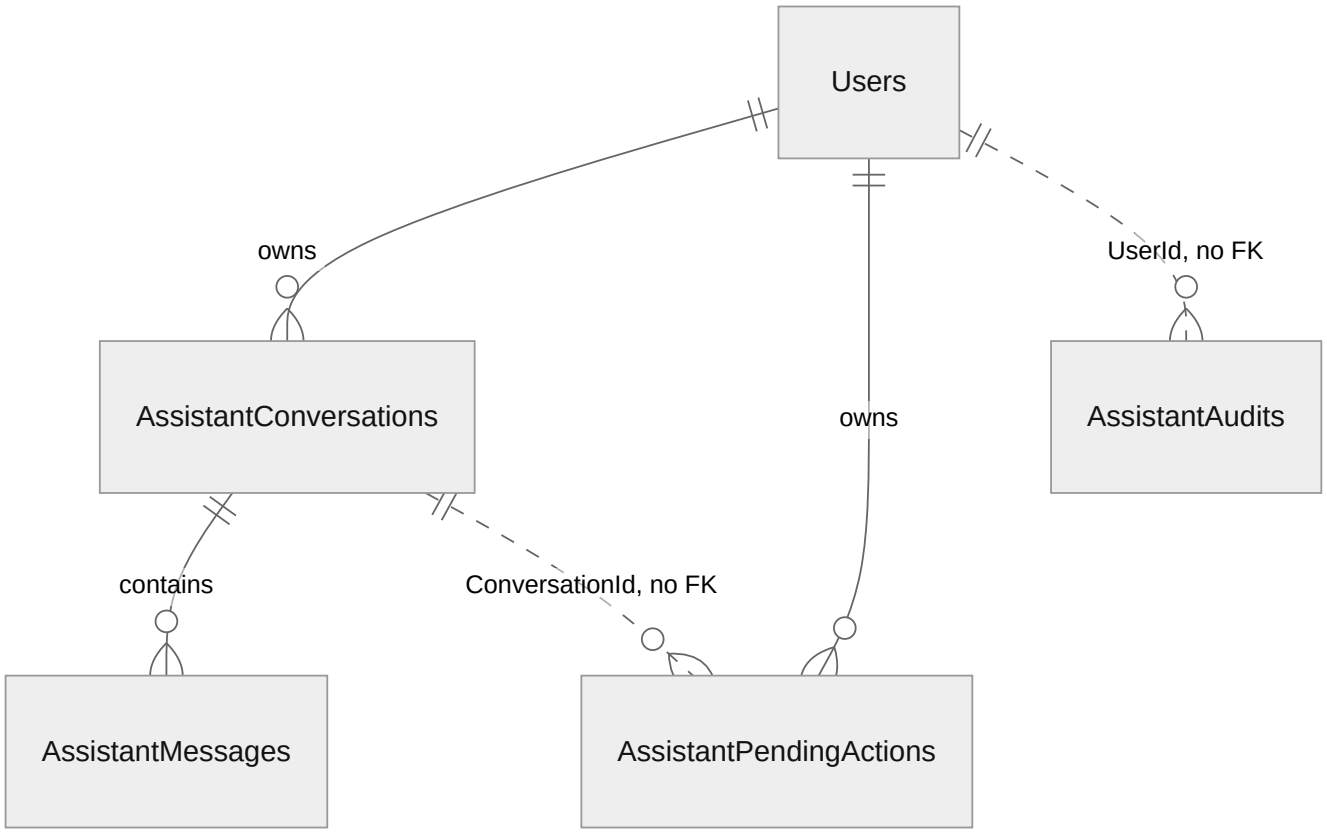


5. Database Design

TFA owns four tables, created by `20261018183210_AddAssistant` (which also creates `AiProviderConfigs`, owned by TFA Settings). Configuration: `Persistence/AppDbContext.cs`.

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5.1 ER Diagram



5.2 Tables

`AssistantConversations` (AssistantConversation)

Column	Type	Null	Key / index	Default	Description
Id	int	No	PK	identity	Conversation id; latest per user is active
UserId	int	No	FK → `Users.Id` (Cascade); `IX_AssistantConversations_UserId`	–	Owner
CreatedAt	DateTime	No	–	–	UTC
UpdatedAt	DateTime	No	–	–	Last message (UTC)

`AssistantMessages` (AssistantMessage)

Column	Type	Null	Key / index	Default	Description
Id	int	No	PK	identity	Message id
ConversationId	int	No	FK → `AssistantConversations.Id` (Cascade); `IX_AssistantMessages_ConversationId`	–	Parent

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Column	Type	Null	Key / index	Default	Description
Role	string(16)	No	–	–	`AssistantMessageRole` as text
Content	string	No	–	–	Message text
MetaJson	string	Yes	–	–	`AssistantMessageMeta` (kind, links, ranked tasks, action id)
ProviderKind	string(16)	Yes	–	–	`AiProviderKind` as text
CreatedAt	DateTime	No	–	–	UTC

`AssistantPendingActions` (AssistantPendingAction)

Column	Type	Null	Key / index	Default	Description
Id	Guid	No	PK	– (set in code)	Action id
UserId	int	No	FK → `Users.Id` (Cascade); `IX_AssistantPendingActions_UserId`	–	Owner
ConversationId	int	No	– (no FK)	–	Conversation it belongs to
Kind	string(32)	No	–	–	`AssistantActionKind` as text
PayloadJson	string	No	–	–	Exact payload shown and executed
Status	string(16)	No	–	– (code sets `Pending`)	`AssistantActionStatus` as text
CreatedAt	DateTime	No	–	–	UTC
ExpiresAt	DateTime	No	–	–	`CreatedAt` + 10 minutes
ResolvedAt	DateTime	Yes	–	–	Set on confirm, cancel, expire, refuse, fail

`AssistantAudits` (AssistantAudit)

Column	Type	Null	Key / index	Default	Description
Id	int	No	PK	identity	Audit id
UserId	int	No	`IX_AssistantAudits_UserId_At` (no FK)	–	Actor
Kind	string(32)	No	–	–	`AssistantActionKind` as text
Outcome	string(32)	No	–	–	prepared / confirmed / cancelled / expired / refused / failed
At	DateTime	No	`IX_AssistantAudits_UserId_At`	–	UTC

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Read by TFA, owned elsewhere: `AiProviderConfigs` (`AiProviderConfig`), `AiModels` (`AiModel`) (TFA Settings); `Users`, `Roles`, `UserDepartments`, `DepartmentManagers` (directory lookups); `Tasks` through `ITaskService`.

5.3 Indexes, Constraints & Triggers

Object	Definition	Purpose
`IX_AssistantConversations_UserId`	(`UserId`)	Latest conversation per user
`IX_AssistantMessages_ConversationId`	(`ConversationId`)	Newest messages
`IX_AssistantPendingActions_UserId`	(`UserId`)	Owner-filtered lookups
`IX_AssistantAudits_UserId_At`	(`UserId`, `At`)	Audit by user and time
`FK_AssistantConversations_Users_UserId`	Cascade	Delete with user
`FK_AssistantMessages_AssistantConversations_ConversationId`	Cascade	Delete with conversation
`FK_AssistantPendingActions_Users_UserId`	Cascade	Delete with user
Enum conversions	`HasConversion<string>()` on `Role`, `ProviderKind`, `Kind`, `Status`	Readable values

5.4 Enumerations

Enum	Values	Stored as
`AssistantActionKind`	SendMail, UpdateOwnProfile, ChangeUserRole	text (32)
`AssistantActionStatus`	Pending, Confirmed, Cancelled, Expired, Refused, Failed	text (16)
`AssistantMessageRole`	User, Assistant, Tool	text (16)
`AiProviderKind`	Ollama, Gemini, OpenAI	text (16) in `AssistantMessages.ProviderKind`
`ProviderFailureKind`	RateLimited, InvalidKey, Timeout, Unreachable, ModelUnavailable, Unexpected	Not stored (`AssistantFailureDto`)
`AssistantMessageKind` (`Dtos/Assistant/AssistantDtos.cs`)	Text, Refusal, Notice	Inside `MetaJson`
`TaskStage` (`Common/Rules/TaskStageCalculator.cs`)	Upcoming, DueToday, Overdue, Completed	Not stored (`PrioritisedTask`)

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5.5 Migrations

Migration	Change
`20261018183210_AddAssistant`	Creates `AssistantConversations`, `AssistantMessages`, `AssistantPendingActions`, `AssistantAudits` (+ `AiProviderConfigs`, TFA Settings) and their indexes

6. Algorithms & Background Jobs

Job / algorithm	Trigger / schedule	Steps	Idempotency
Tool loop (`AssistantService.RunTurnAsync`)	Each message	≤ 4 rounds; tools advertised per caller; fixed reply ends the turn; tool output redacted	Nothing stored until the provider answers
History window (`LoadHistoryAsync`)	Each message	24 newest → drop notices → 12 → ≤ 6000 chars → chronological	Read only
Secret screen (`SecretScreen`)	Each message; each tool result	Regex patterns: password phrases, `sk-`, `AIza`, `Bearer`, JWT, long mixed token	Pure function
Ranking (`TaskPrioritiser.Rank`)	`prioritise_tasks`	Stage → local due date → priority desc → deadline → task code	Deterministic, provider-independent
Help search (`HelpKnowledgeBase.Search`)	`search_help`	Score ≥ 3, top 3, role redaction	Pure function over embedded topics
Confirm once (`TryClaimAsync`)	`POST …/confirm`	`UPDATE … SET Status = 'Confirmed' WHERE Id = @id AND Status = 'Pending'`	Only one concurrent confirm wins
Rate limit (`AssistantRateLimiter`)	Each counted message	Sliding 1-minute window, 20 per user	In memory, per process; reset on restart
Key-ring check (`KeyRingProtectionCheck.Check`)	Application start (`Program.cs`)	Warns when the key ring is unprotected or cannot be checked	Read only

Pending-action expiry is lazy (on read or confirm); no background job belongs to this sub-module.

7. Security Design (detailed)

Item	Implementation
Identity	`AssistantCaller` from validated claims only (`ClaimTypes.NameIdentifier`, `ClaimTypes.Role`); never from chat text or model output

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Item	Implementation
Permission checks	`perm:assistant.chat` on the 4 chat endpoints; `perm:assistant.actions.confirm` on the 3 action endpoints; policies registered for every `PermissionCatalog.All` entry (`RequireClaim(AuthClaims.Permission, key)`); tools and confirms evaluated with the same policies via `AssistantAuthorizer`; `propose_role_change` and the `ChangeUserRole` confirm need `perm:roles.manage`
Defaults	`assistant.chat`, `assistant.actions.confirm`: `GrantedToManager: true, GrantedToEmployee: true, GrantToAllExistingRoles: true` (`PermissionCatalog.cs`); Administrator holds all assignable keys
Data scope	Conversations and actions filtered by `UserId` (another user's id → 404, also for Administrators); task reads through `ITaskService` with the caller's scope; people lookups active users only
Model containment	Only advertised tools callable; action tools only create pending actions; confirm only via the user's HTTP call; refusals and confirmations are fixed text (`AssistantTexts`); tool output marked as data and redacted
Secrets	`SecretScreen` blocks passwords/keys in user text; provider API keys encrypted with Data Protection purpose `TaskFlow.AiProviderKeys` (key ring `AiProviders:KeyRingPath`, DPAPI-protected on Windows), decrypted in memory for one call; provider `HttpClient`s without logging handlers, `AllowAutoRedirect = false`
Cloud notice	Client holds the first message to a Cloud provider until **I understand** (`tfa.cloudAck`)
Audit	`AssistantAudits` (no bodies, passwords or keys); endpoint 403s recorded by `DeniedActionAuditHandler`
Frontend	Panel rendered only when `canUseAssistant()` (`Permissions.AssistantChat`); action buttons not gated by `assistant.actions.confirm` (§10)

8. Configuration

Key	Default	Purpose
`AiProviders:KeyRingPath`	`<secret>` (omitted from the fact sheet; `dp-keys` per module doc)	Data Protection key ring folder (relative to `Storage:Root`)
`AiProviders:TimeoutSeconds`	`60`	`HttpClient.Timeout` of every provider call
`Storage:Root`	`""`	Base folder for the key ring
`Application:Timezone`	`Asia/Kolkata`	Fallback time zone for "today" / overdue (`IAppClock`)
`ConnectionStrings:Default`	`<secret>`	SQLite database
`AssistantRules.*` (constants)	2000 / 12 / 6000 / 4 / 100	Message length, history messages, history characters, tool rounds, loaded messages
`AssistantActionRules.Lifetime` (constant)	10 minutes	Pending-action lifetime

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Key	Default	Purpose
`AssistantRateLimiter.RequestsPerMinute` (constant)	20	Messages per user per minute
`HelpKnowledgeBase.MinimumScore` (constant)	3	Help match threshold
Frontend `localStorage`	`tfa.provider`, `tfa.cloudAck`, `tfa.handsfree`	Chosen model, cloud acknowledgement, deliberate voice off
Frontend constants	`PROVIDER_REFRESH_MS` 60000, `COMMAND_SILENCE_MS` 2000, `WAKE_CAPTURE_TIMEOUT_MS` 8000, `DICTATION_TIMEOUT_MS` 10000, `MAX_SPOKEN_CHARS` 500	Panel and voice timing

Built-in default models (seeded by TFA Settings): `llama3.2:3b`, `gemini-2.0-flash`, `gpt-4o-mini`.

9. Testing Design

Area	Test class / file	Type	Covers
Endpoints	`TaskFlow.IntegrationTests/Assistant/AssistantEndpointsTests.cs`	Integration	7 endpoints, policies, owner filter
Providers list	`TaskFlow.IntegrationTests/Settings/TfaSettings/AssistantProviderEndpointsTests.cs`, `AiDefaultsOnFirstRunTests.cs`	Integration	Provider options, default models
Key ring	`TaskFlow.IntegrationTests/Common/KeyRingStartupWarningTests.cs`	Integration	Startup warning
Chat engine	`TaskFlow.UnitTests/Assistant/Chat/AssistantServiceTests.cs` (+ `AssistantServiceHarness.cs`)	Unit	Length, history, tool loop, persistence
Gateway	`TaskFlow.UnitTests/Assistant/ModelGateway/ChatModelGatewayTests.cs`, `ProviderFailureHandlingTests.cs`	Unit	Dispatch, failure classification
Secrets	`TaskFlow.UnitTests/Assistant/Rules/SecretScreenTests.cs`	Unit	Detection and redaction
Ranking	`TaskFlow.UnitTests/Assistant/Rules/TaskPrioritiserTests.cs`	Unit	Work order
Actions	`TaskFlow.UnitTests/Assistant/Actions/AssistantActionServiceTests.cs`, `AssistantActionConcurrencyTests.cs` (+ `AssistantActionHarness.cs`)	Unit	Lifecycle, confirm once
Tools	`TaskFlow.UnitTests/Assistant/Tools/AssistantReadToolsTests.cs`, `AssistantActionToolsTests.cs`	Unit	Read and action tools

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Area	Test class / file	Type	Covers
Gate	`TaskFlow.UnitTests/Assistant/Toolbox/AssistantToolGateTests.cs`	Unit	Tool authorization
Help	`TaskFlow.UnitTests/Assistant/HelpKnowledge/HelpKnowledgeBaseTests.cs`	Unit	Search and redaction
Key protection	`TaskFlow.UnitTests/Integrations/Ai/KeyProtection/AiKeyProtectorTests.cs`, `KeyRingProtectionCheckTests.cs`	Unit	Encryption, key-ring check
Provider config	`TaskFlow.UnitTests/Settings/TfaSettings/ProviderConfig/AiProviderConfigServiceTests.cs`, `AiProviderSeederTests.cs`	Unit	Config used by TFA
Architecture	`TaskFlow.UnitTests/Architecture/ArchitectureTests.cs`	Unit	Layer rules

> ⚠️ To be confirmed: coverage is inferred from file names only. No dedicated test file exists for `AssistantRateLimiter` or `AssistantAuditLog`, and no Angular `*.spec.ts` exists for `modules/assistant`.

10. Known Limitations & Open Issues

- **Role change bypasses Employee Management rules:** `propose_role_change` / `ConfirmAsync` check only `roles.manage` and call `IEmployeeService.UpdateAsync`; `PUT /api/v1/users/{id}` also requires `employees.role.assign` (and `roles.administrator.assign` for the Administrator role). ⚠️ To be confirmed.
- **Confirm buttons not gated:** `TfaActionCardComponent` shows Confirm / Edit / Cancel without checking `assistant.actions.confirm`; the click returns 403.
- **"My manager"** finds only users with `RoleId == SystemRoles.ManagerId`; department managers in custom roles are ignored; the `AssistantManagerResolver` doc comment still says "project".
- **Help redaction by role name:** a custom role always gets the `others` note for restricted topics.
- **Texts out of date:** `AssistantTexts.PasswordGuidance` and `change-password.md` mention the current password and "other sessions"; the real flow uses an authenticator code and ends every session (LLD-My-Profile 4.1.2).
- **Two "not available" texts:** server 400 "TFA is not available yet. An administrator needs to configure an AI provider."; panel empty state "An administrator needs to choose an AI model in TFA Settings."
- **Lazy expiry:** no job marks old actions `Expired`; the card shows a fixed "Expires in 10 minutes".
- **No retention:** conversations and messages are kept indefinitely; no "delete my history". ⚠️ To be confirmed.
- **`ProviderNotConfiguredException`** is not an `AppException`; a provider disabled between the availability check and the call surfaces as 500.
- **Rate limiter and audit** are not covered by a dedicated test file; the limiter is per process.
- ⚠️ To be confirmed: whether `IMailService.SendAsync` raises a mail notification for recipients (Mail module).

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11. Traceability (FRD → code)

FRD requirement	Endpoint / class	Test
FR-TFA-01	`shell.component.html` `canUseAssistant()` · `perm:assistant.chat` on 4 endpoints	`AssistantEndpointsTests`
FR-TFA-02	`TfaPanelComponent`, `AssistantService.open` / `close` / `toggle` (Angular)	⚠️ To be confirmed (no frontend spec)
FR-TFA-03	`GET /api/v1/assistant/conversation` (`ConversationMessages` 100) · `GET /api/v1/assistant/providers`	`AssistantEndpointsTests`
FR-TFA-04	`TfaComposerComponent` `TFA_SUGGESTIONS`	⚠️ To be confirmed (no frontend spec)
FR-TFA-05	`TfaComposerComponent` (Enter / Shift+Enter)	⚠️ To be confirmed (no frontend spec)
FR-TFA-06	`FindLatestForUpdateAsync(userId)` · `FindForUserAsync(id, userId)` · `RunActionAsync` 404	`AssistantEndpointsTests`
FR-TFA-07	`POST /api/v1/assistant/conversation/messages` · `AssistantRules.MaxMessageLength` · `AssistantRequestException`	`AssistantServiceTests`
FR-TFA-08	`SecretScreen.ContainsSecret` · `AssistantTexts.SecretBlocked`	`SecretScreenTests`
FR-TFA-09	`GetMyTasksTool` · `AssistantTaskReader` · `ITaskService.ListAsync`	`AssistantReadToolsTests`
FR-TFA-10	`TaskPrioritiser.Rank`	`TaskPrioritiserTests`
FR-TFA-11	`PrioritiseTasksTool` · `PrioritisedTask` · `TfaMessageComponent`	`TaskPrioritiserTests`, `AssistantReadToolsTests`
FR-TFA-12	`GET /api/v1/assistant/providers` · `IAiProviderConfigService.GetAvailableAsync` · `TfaHeaderComponent` · `tfa.provider`	`ChatModelGatewayTests`, `AssistantProviderEndpointsTests`
FR-TFA-13	`AssistantService.SendAsync` (no provider) · `ProviderFailureMessages.NotConfigured` · panel empty state	`AssistantServiceTests`
FR-TFA-14	`AssistantMessage.ProviderKind` · `AssistantMessageDto.Provider`	`AssistantServiceTests`
FR-TFA-15	`AssistantService.acknowledgeCloud` (Angular) · `tfa.cloudAck`	⚠️ To be confirmed (no frontend spec)
FR-TFA-16	`ChatModelGateway` · `ProviderFailureKind` · `AssistantFailureDto` · `ProviderFailureMessages.For` · `AssistantService.retry`	`ProviderFailureHandlingTests`, `ChatModelGatewayTests`
FR-TFA-17	`POST /api/v1/assistant/conversation/new` · `NewConversationAsync`	`AssistantEndpointsTests`

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FRD requirement	Endpoint / class	Test
FR-TFA-18	`LoadHistoryAsync` · `HistoryMessages`, `HistoryCharacters`	`AssistantServiceTests`
FR-TFA-19	`RunTurnAsync` · `MaxToolRounds`	`AssistantServiceTests`
FR-TFA-20	`SecretScreen.Redact` on tool output · `AssistantPrompt`	`AssistantServiceTests`
FR-TFA-21	`AssistantService.loadConversation` on user change (Angular)	⚠️ To be confirmed (no frontend spec)
FR-TFA-22	`AssistantPrompt.Build` · `IAppClock.TodayLocalAsync` · `TaskPrioritiser`	`AssistantReadToolsTests`
FR-TFA-23	`AssistantTexts.PermissionDenied` · `AssistantMessageKind.Refusal`	`AssistantToolGateTests`
FR-TFA-24	`tfa-panel.component.html` footer	⚠️ To be confirmed (no frontend spec)
FR-TFA-25	`HelpKnowledgeBase.Search` (`MinimumScore` 3, `maxResults` 3) · 19 embedded topics	`HelpKnowledgeBaseTests`
FR-TFA-26	Topic `link` → `AssistantLinkDto` · `TfaMessageComponent`	`HelpKnowledgeBaseTests`
FR-TFA-27	`HelpKnowledgeBase.ToDto` (`IsRestricted`, `others`)	`HelpKnowledgeBaseTests`
FR-TFA-28	Empty `search_help` result · `AssistantPrompt` rules	`HelpKnowledgeBaseTests`
FR-TFA-29	`AssistantActionService.CreateAsync` · `AssistantTexts.ActionPrepared` · `ToolOutcome.FixedReply`	`AssistantActionServiceTests`
FR-TFA-30	`AssistantToolGate` · `AssistantActionRules.Requirement` · `IAssistantAuthorizer.IsAllowedAsync` in `ConfirmAsync`	`AssistantToolGateTests`, `AssistantActionServiceTests`
FR-TFA-31	`AssistantActionRules.Lifetime` · `ConfirmAsync` expiry · `SetStatusAsync(Expired)`	`AssistantActionServiceTests`
FR-TFA-32	`POST /api/v1/assistant/actions/{id}/confirm` · `TryClaimAsync`	`AssistantActionConcurrencyTests`
FR-TFA-33	`POST /api/v1/assistant/actions/{id}/cancel` · `CancelAsync`	`AssistantActionServiceTests`
FR-TFA-34	`DraftMailTool` · `SendMailRequestValidator` · `IMailService.SendAsync`	`AssistantActionToolsTests`
FR-TFA-35	`AssistantManagerResolver.ResolveAsync` · `GetActiveManagersOfDepartmentsAsync`	`AssistantActionToolsTests`
FR-TFA-36	`AssistantManagerResolver` (`ITaskService.GetDetailsAsync` null)	`AssistantActionToolsTests`
FR-TFA-37	`PUT /api/v1/assistant/actions/{id}` · `EditMailAsync` · `ResolveRecipientsAsync`	`AssistantActionServiceTests`
FR-TFA-38	`EnsurePending`	`AssistantActionServiceTests`

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FRD requirement	Endpoint / class	Test
FR-TFA-39	`ProposeProfileUpdateTool` · `ExecuteAsync(UpdateOwnProfile)` → `IProfileService.UpdateAsync`	`AssistantActionToolsTests`
FR-TFA-40	`ProposeRoleChangeTool` (`Policy` `perm:roles.manage`, self refusal, exact role)	`AssistantActionToolsTests`, `AssistantToolGateTests`
FR-TFA-41	`ExecuteAsync(ChangeUserRole)` → `IEmployeeService.UpdateAsync` (stamp rotation, token revoke)	`AssistantActionServiceTests`
FR-TFA-42	`FindUserTool` · `IMailService.GetContactsAsync` (max 10)	`AssistantReadToolsTests`
FR-TFA-43	`ExplainPasswordChangeTool` · `AssistantTexts.PasswordGuidance`	`AssistantReadToolsTests`
FR-TFA-44	`ContactAdminGuidanceTool` · `AssistantTexts.AdminGuidance`	`AssistantReadToolsTests`
FR-TFA-45	`ConfirmAsync` failure branch (`Failed`, "I could not do that: ...")	`AssistantActionServiceTests`
FR-TFA-46	`AssistantService.confirm` / `cancel` / `edit` fallbacks (Angular)	⚠️ To be confirmed (no frontend spec)
FR-TFA-47	`WakePhraseService` (`greet`) · `TfaPanelComponent` · `SpeakService`	⚠️ To be confirmed (no frontend spec)
FR-TFA-48	`WakePhraseService` (`interim`, `question`, `COMMAND_SILENCE_MS`)	⚠️ To be confirmed (no frontend spec)
FR-TFA-49	`TfaComposerComponent` mic · `DICTATION_TIMEOUT_MS`	⚠️ To be confirmed (no frontend spec)
FR-TFA-50	`TfaPanelComponent.answerAloud` · `MAX_SPOKEN_CHARS`	⚠️ To be confirmed (no frontend spec)
FR-TFA-51	`TFA_VOICE_KEY` (`tfa.handsfree`) · `WakePhraseService` retry/back-off	⚠️ To be confirmed (no frontend spec)
FR-TFA-52	`SPEECH_RECOGNITION_FACTORY` · error messages	⚠️ To be confirmed (no frontend spec)
FR-TFA-53	`AssistantRateLimiter.TryAcquire` · `AssistantRateLimitedException`	⚠️ To be confirmed (`AssistantServiceTests` by name only)
FR-TFA-54	`AssistantAuditLog.RecordAsync` · `AssistantAudits`	⚠️ To be confirmed (`AssistantActionServiceTests` by name only)
FR-TFA-55	`AssistantAudit` columns (`UserId`, `Kind`, `Outcome`, `At` only)	⚠️ To be confirmed
FR-TFA-56	`AssistantToolbox.ExecuteAsync` refusal audit · `ConfirmAsync` refusal audit	⚠️ To be confirmed (`AssistantToolGateTests` by name only)

Coverage: 56 of 56 `FR-TFA-nn` traced; 7 of 7 `AssistantController` actions documented.

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